VRIJE UNIVERSITEIT BRUSSEL

SINGLE TRIAL EEG CLASSIFICATION TO PREDICT HUMAN BRAIN STATES

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sciences and bioengineering sciences

Abstract

Studying electric and magnetic activity which is produced by human brain while at rest or during performing the tasks helps to understand working of human brain in a different kind of subjects such as normal subjects or subjects with certain mental illness. We can use a machine learning techniques to determine on the basis of recorded signals, whether the subject is experiencing an expected or an unexpected stimulus. By studying brain activity, we can determine whether the subject is experiencing an mental illness or it is a normal subject by analysing different electric activity of brain. We can use a EEG for recording electric and magnetic activity of brain. The ability to effectively analyse and classify EEG is the foundation for building usable Brain-Computer Interfaces as well as improving the performance of EEG analysis software which is used in clinical and research settings.

In this thesis, we work with Auditory P300 paradigm in which subject hears the tones such as standard, target, and distractor with frequencies 1000 Hz, 1500 Hz, and 500 Hz respectively. We used a machine learning(ML) techniques for single-trial classification of target and distractor from EEG data for normal and schizophrenia subjects. This will helps to determine difference in brain signals for different stimuli in normal and schizophrenia brain. The P300 signal is an event related potential (ERP) which means that the signal is seen on an EEG as a rapid single potential change as a response to a sensory and cognitive event. The signal's peak comes an average of 300 milliseconds after the stimulus so we call it the P300. To build a ML techniques for this classification we used traditional ERP features which consists of signal peak maximum amplitude and latency and brain connectivity features which consists of various network measures.

EEG data used in this thesis consist of 54 normal and 54 schizophrenia subjects. The EEG-data were high-pass and low-pass filtered, epoched and averaged. ERP features such as peak amplitude and latency were extracted from the averaged signals. Connectivity features are extracted from correlation matrices of each trials using Brain Connectivity Toolbox of MATLAB. The resulting datasets from ERP features and connectivity features was used to train and test classification algorithm, First on single normal and schizophrenia subject, Second on group of 10 normal and schizophrenia subject, and Third on group of 54 normal and schizophrenia subjects. For ML techniques, we used nested cross validation for simulatneous processing of feature selection, feature preprocesssing, model selection, and hyperparameter tuning. We then used logistic regression, random forest, decision tree, K-nearest neighbors, and SVM classifiers using nested cross validation to train the model.

We trained the model for ERP features and connectivity features. We observed that accuracy for single-trial classification with the connectivity features increased in comparison with ERP features for the group of 54 normal and schizophrenia subject.

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Contents

1 Introduction

The fundamental goal of cognitive neuroscience is understanding the basic neural processes that underlie complex higher-order cognitive operations and functional domains. An electroencephalogram (EEG) is a clinical and relatively inexpensive method for assessing the neurophysiological function that can be used to achieve this goal. EEG is a technique that records the electrical activity of large populations of neurons in the brain by placing an array of electrodes on the surface of a subject's scalp. EEG provides a temporally correct measure of neurophysiological function during different tasks and conditions. There are wide-ranging applications of this method since it allows researchers to explore the infinite domains in which it is important to understand the electric activity of neurons in a brain. [23].

Brain electric and magnetic activity or neural signals obtained by EEG recordings plays an important role in medical, health applications and in Brain Computer Interface (BCI) [21]. BCI which is also referred to as a brain-machine interface (BMI) or neural interface, is a system that establishes direct communication between the brain and an external device by mean of brain-activity which is being generally measured by EEG. The basic setup of a BCI system includes three components[32]:

- **Specific electrodes** to record electric and magnetic brain activity,
- **A processing pipeline** to interpret those signals which extract relevant features from them, decodes patterns of interest, and outputs commands.
- **A computer or external device** which operates via the generated commands.

The biggest challenge for BCI's is that every individual has different brain. Using Machine Learning, for every new session, the BCI has to learn from the individuals brain activity. EEG is one effective way of recording and analyzing the brain activity due to its ease of use, affordable cost and fine resolution. Neurons in a brain get fired due to brain activity producing higher electrical potential which is recorded by the electrodes attached to the scalp of a human being.

Event-related potential (ERP) experimental design is used by many EEG researchers in which a large number of experimental trials are averaged together. ERP is a measured brain response obtained from unexpected, specific sensory, or cognitive event. P300 is an electric ERP signal which is seen on an EEG as a rapid single potential change as a response to a sensory, cognitive, or unexpected stimuli. We call it a P300 because the signal's peak comes an average of 300 milliseconds after, or ”post”, the stimulus. The P300 signals are attention related which requires a response to their “target” stimulus. The oddball paradigm is a frequently used paradigm where low probability target events are mixed with high probability standard events. The P300 is observed following the target events[27].

Another important properties for classification of brain signals obtained with EEG is connectivity between different brain regions. Brain connectivity describes the network of functional connections across brain. A network is defined as a collection of nodes and links between pairs of nodes. In brain network, brain regions are represented as a nodes and links represents anatomical, functional, or effective connections. All networks can be represented with connectivity matrices where rows and columns represents nodes and entries denotes links. Brain networks can be characterize using complex network analysis approach with small number of neurobiologically meaningful and easily computable measures since brain networks are invariably complex and shares a number of common features with a network from other biological or physical system. Complex network analysis is based on mathematical study of network which is known as graph theory. Network characterization of structural and functional connectivity data are described by a network measures. Network measures detects aspects of functional integration and segregation, measures the quantity of importance of individual brain regions, and characterize patterns of local anatomical circuitry in a several ways[30].

In this thesis, brain connectivity features are also extracted from EEG data for training a ML model for classification of target and distractor trial. For connectivity features,the most widely used and important network measures are considered, which are obtained using the matlab based brain connectivity toolbox (http://www.brain-connectivity-toolbox.net). One of a basic and important measure of a network is degree of a node which is a number of links connected to that node or number of neighbors of that nodes. Degree distribution is degrees of all nodes in a network. Mean network degree is commonly used as a measure of density or wiring cost of network. Weighted variant is sum of all neighboring link weights. The fraction of triangles around an individual node is known as the clustering coefficient and a classical variant of the clustering coefficient is known as transitivity. The most commonly used measure of functional integration is characteristic path length of the network and it is average shortest path length between all pairs of nodes in the network. Betweenness centrality is a measure of centrality which is defined as the fraction of all shortest paths in the network that pass through a given node. The closeness centrality is a inverse of average shortest path length from one node to all other nodes in network. The average inverse shortest path length is a related measure known as global efficiency. A modularity is a statistic which assess the degree to which the network may be subdivided into such non-overlapping groups. These network measures are briefly described in upcoming chapter.

The goal of this thesis is to consider two feature sets: ERP features and connectivity features for training a ML classifier for single-trial EEG classification and compare the accuracy with these two feature sets to determine whether connectivity features are better/worse than a conventional ERP features.

1.1 Organization of the thesis

After introduction, second section is about background which briefly explains about Electroencephalography (EEG), brain connectivity and network, and brain network measures. Third section is about method which briefly explains machine learning techniques, data, feature detection, and dataset preparation for ML techniques. Fourth section is about results which briefly shows the results obtained after the different experiments and performances with ERP and connectivity features. The next section is about discussion where we discussed about performance with two feature sets, feature importance, limitations, and future work. Finally in section 6, this thesis is concluded with the study observed during the different experiments.

2 Background

To help understanding the further discussion by the readers, basic structure and function of human brain, EEG and brain connectivity are explained under this section.

2.1 Electroencephalography (EEG)

An electroencephalogram (EEG) is a test that detects electrical activity in human brain using electrodes which consists of small metal discs with thin wires attached to the scalp. Your brain cells communicate via electrical impulses and are active all the time, even when you're asleep. This activity shows up as wavy lines on an EEG recording. The electrodes detect tiny electrical charges that result from the activity of the brain cells. The charges are amplified and appear as a graph on a computer screen, or as a recording that may be printed out on paper. An EEG is one of the main diagnostic tests for brain disorders such as epilepsy and sleep disorders.

EEG signals are short-time and periodic and the periodic activity is distributed into different frequency bands. Different people of different ages may have different amplitude and frequency of EEG signals while they are recorded in different states such as performing a task or relaxing. Based on frequency ranges, five types of waves can be identified. They are alpha , theta, beta, delta and gamma from low to high frequency respectively [34]. Different mental states are associated with different waves. Each brainwave has a distinct purpose and helps us behave, think, move and process.

- **Delta Waves** Delta waves are associated with deep levels of relaxation. Delta waves are created by brain when individual is in healthful and deep sleep. Higher levels of delta waves are more commonly found in young children and they are the slowest recorded brain waves in humans. During the aging process, lower Delta waves are produced.

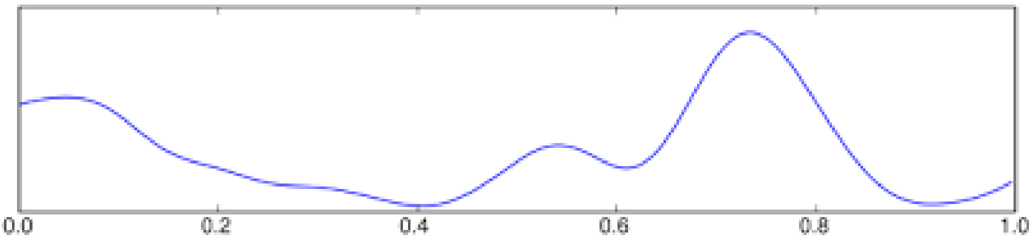

Figure 1: Delta wave. This image is retrieved from WikiDoc[7].

- **Theta Waves** Theta waves are flexible waves because of their prevalence when one is in a dream or hypnotic state. Theta waves are commonly found when one is in the state of daydreaming or asleep. In the state of dream or hypnotic state, a brain's theta waves are optimal and one is more prone to hypnosis and associated therapy.

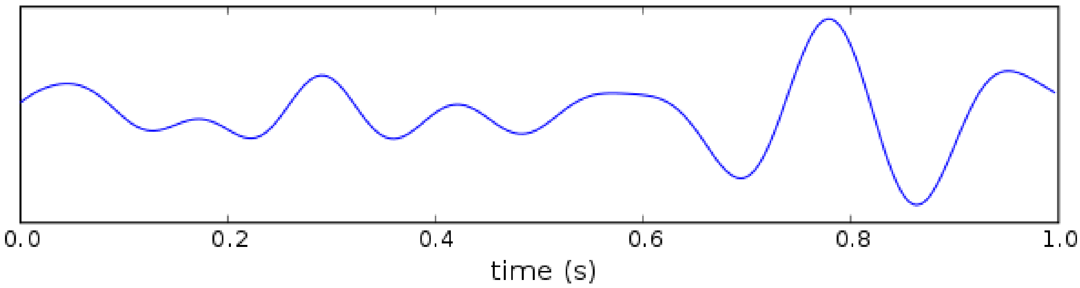

Figure 2: Theta wave. This image is retrieved from Wikipedia[31].

- **Alpha Waves** Alpha waves are the 'frequency bridge' between our conscious thinking and subconscious mind. They helps to play a active role in network coordination and communication, to calm you down and promote feelings of deeper relaxation.

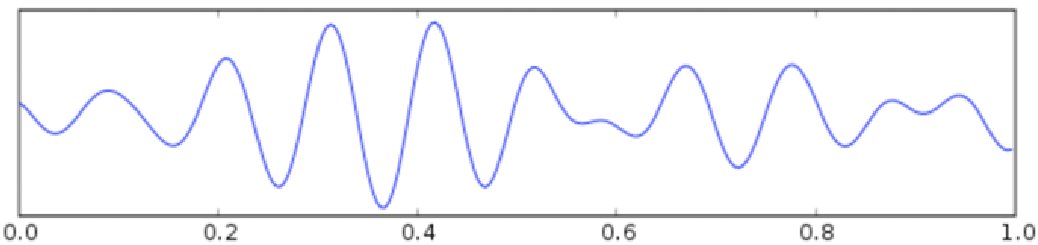

Figure 3: Alpha wave [28].

- **Beta Waves** Beta waves are the high frequency waves and they are most commonly found in awake state of humans. They are occurred during conscious states such as cognitive reasoning, writing, reading, speaking or thinking.

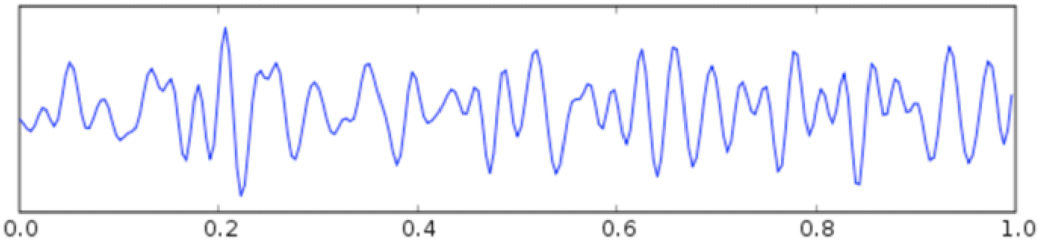

Figure 4: Beta waves [28].

- **Gamma Waves** In addition to healthy cognitive reasoning, Gamma waves are created during the process of more complex tasks such as learning, or a critical reasoning.

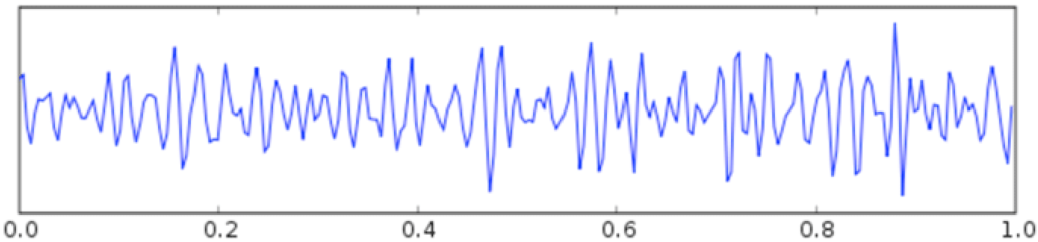

Figure 5: Gamma waves [28].

2.1.1 10-20 System of Electrodes Placement[4]

10-20 System is one of commonly used methods of electrode placement for recording EEG signals which is standardized by the American Electroencephalographic Society. This system uses distances in percentages from a couple of fixed points on the head. The starting points are the nasion, at the top of the nose, and the inion which is the boney bump at the back of the skull. There is an imaginary vertical line from the nasion to the inion, and a horizontal line from the left ear lobe to the right. Parameters of skull are measured from above points. Locations for electrodes are determined by division of parameters to intervals of 10% and 20%. Three electrodes are placed on the middle of adjacent points. Relationship between the location of an electrode and cerebral cortex is the basis of 10-20 system. A combination of letter(s) and an integer number is used to determine the placement.

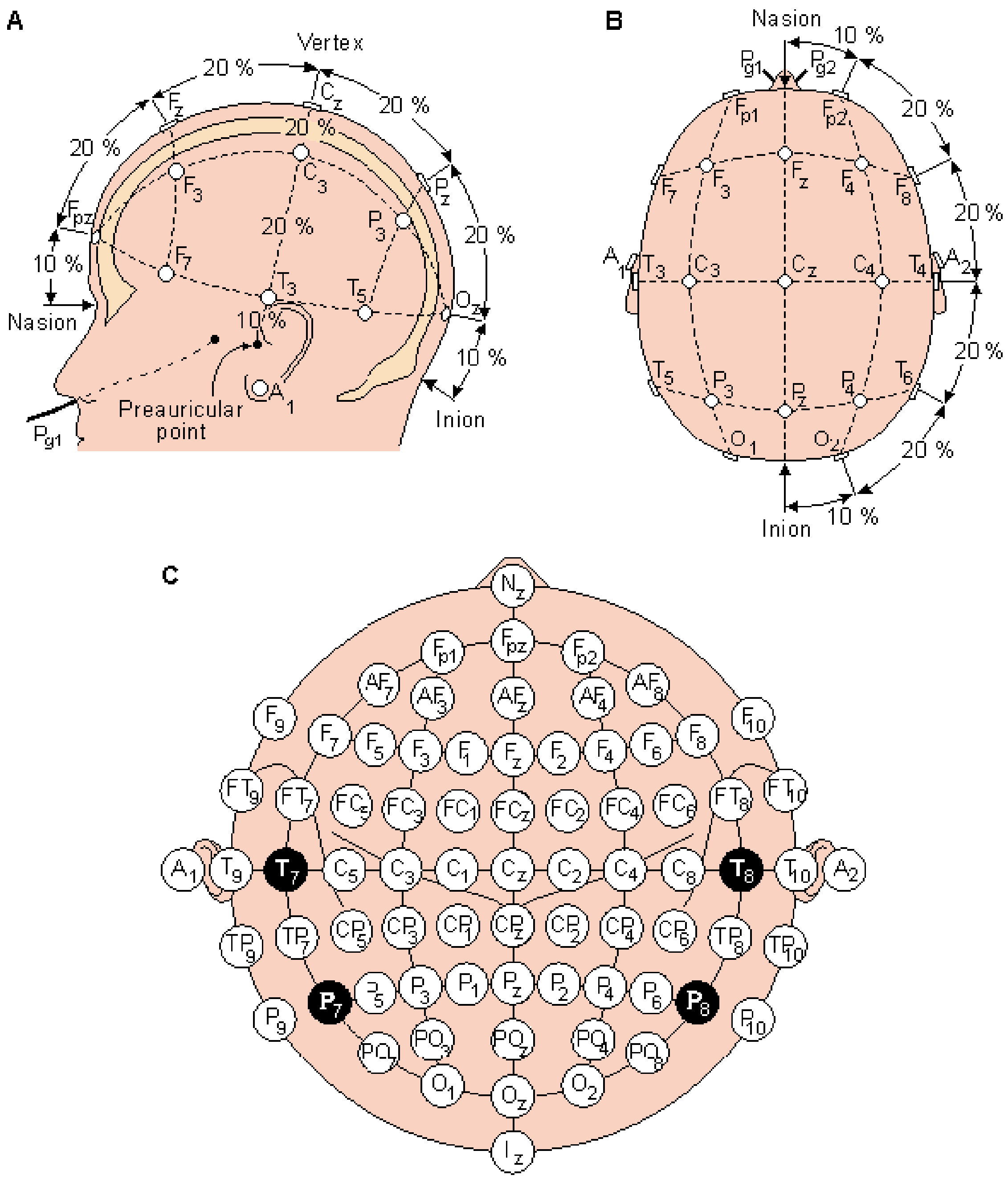

Figure 6: Electrode placement in EEG. This image is retrieved from 'User Tutorial:EEG Measurement Setup'

2.1.2 EEGLAB

EEGLAB is a toolbox and graphical user interface that runs under MATLAB which processes continuous and event-related EEG, MEG and other electrophysiological data. EEGLAB can process collection of EEG data for any number of channels. The various functions of EEG data processing such as importing channels, event information, and visualization of data is supported by EEGLAB. Other functions supported by EEGLAB are multi-trial ERP-image plots, scalp map and dipole, model plotting, scrolling, data pre-processing, independent component analysis (ICA) and time/frequency decompositions including channel and component cross-coherence supported by bootstrap statistical methods based on data resampling. Data preprocessing includes filtering, epoch selection, averaging and artifact rejection. To divide the functionalities of EEGLAB, three layers are designed. First layer allows the user to interact with data using graphical interface without using any syntax of MATLAB. According to memory available, user can also mold the operation of MATLAB. Under the middle layer, operations such as data processing, command history, and interactive pop functions can be performed. Individual signal processing function and EEGLAB data structures can be used to write custom scripts by experienced users of MATLAB [28].

We can process EEG and other brain data flexibly and interactively because EEGLAB provides an interactive graphic user interface. EEGLAB interactively process these data using independent component analysis, time or frequency analysis and standard averaging methods[33].

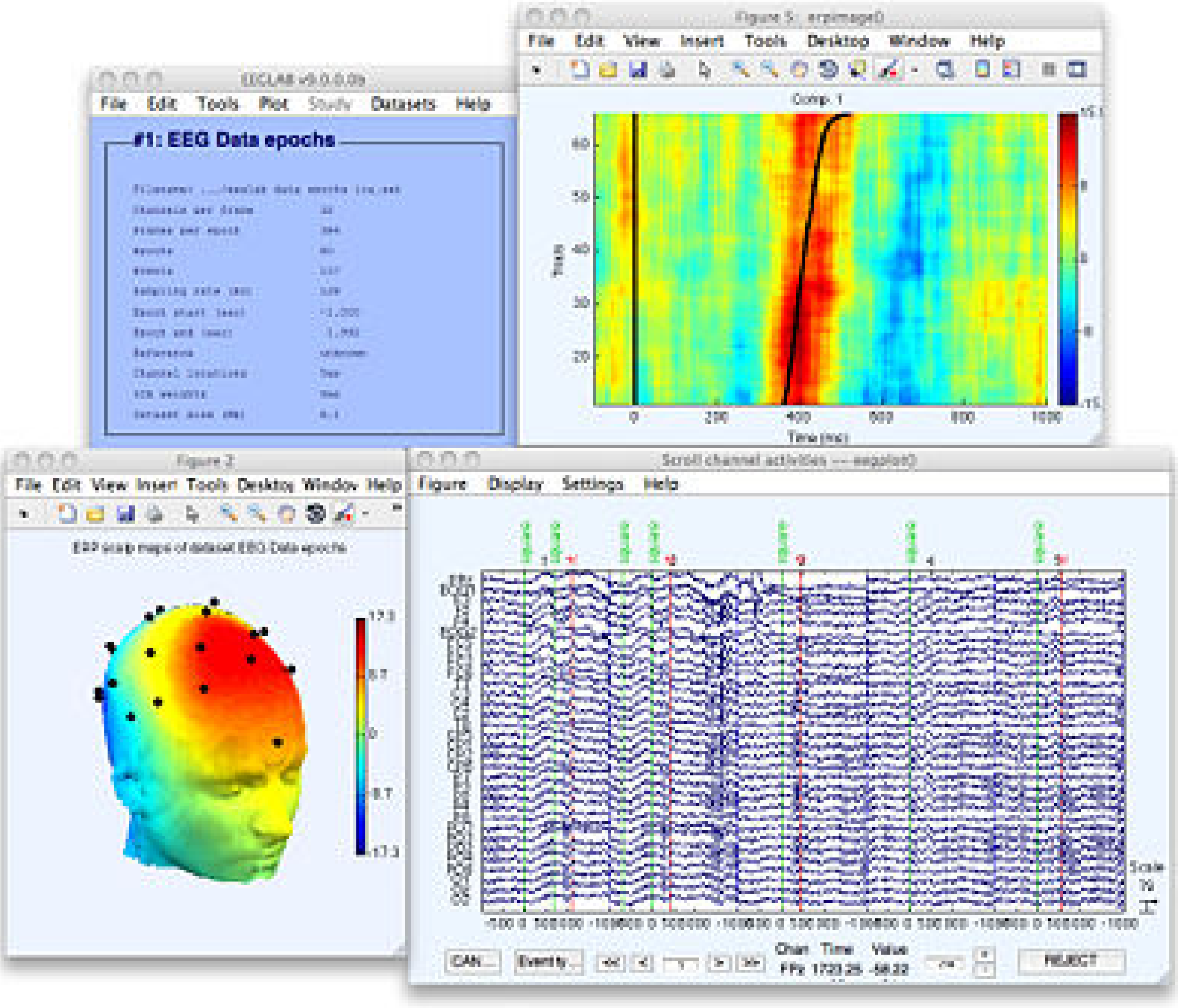

Figure 7: EEGLAB Toolbox [33]

2.2 Brain connectivity and networks

2.2.1 Brain

The human brain is the central and complex organ of the human nervous system, and with the spinal cord makes up the central nervous system. The brain consists of the cerebrum, the brainstem and the cerebellum. It handles all kind of physical tasks and controls almost all the core activities required for survival of human. Human brain contains 5 to 10 billion neurons approximately, which is connected by as many as 100 billion synapses. Neurons are the basic information processing unit which are connected together for flow of information. The human brain has information memory and storage functions. Human memory is controlled by a very rich variety of biological signals, which is the precipitation of information and knowledge by individuals [8].

The different parts of human brain is as shown in below figure. Spinal cord and large brain are connected through brainstem. The human brain is divided into three different parts based on its functionality. First part is called as forebrain. It is also known as cerebrum or large brain which handles high level mental tasks such as computational thinking. Second part of brain is called as brainstem and it controls the vital functions such as cardiac and respiratory functions and third part of brain is called cerebellum and it is responsible for body movement [28].

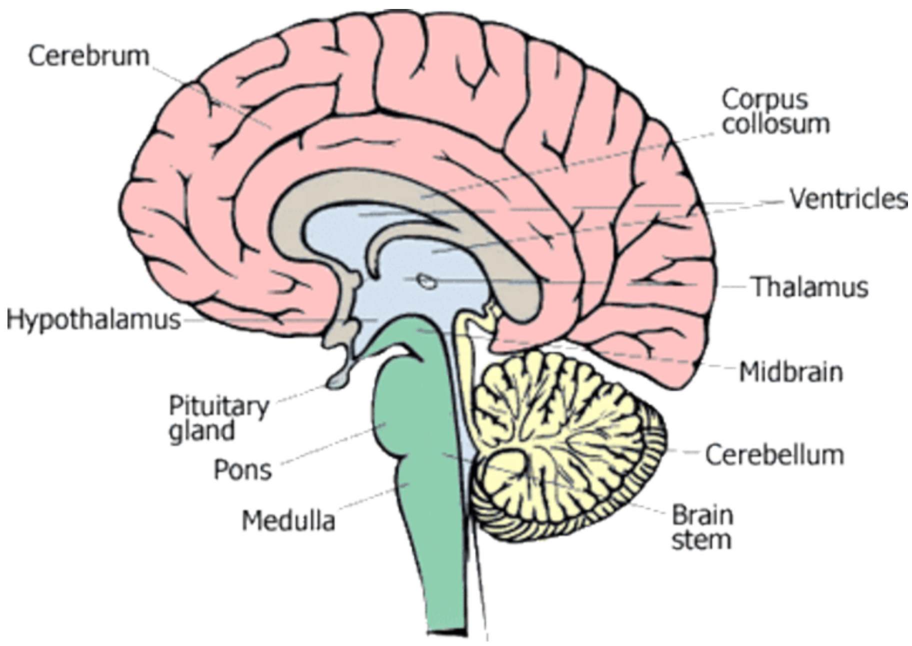

Figure 8: Different parts of human brain [28]

2.2.2 Brain network

It is a set of interconnected brain regions with the information transfer among the regions. The brain is characterized by miscellaneous patterns of structural connections which support cognition and a variety of behaviors. The brain network system consists of collection neurons or nerve cells. The primary goal of neuron network is to transmit and process information received from our senses. To construct a brain network we define a set of nodes which is a region of interest of brain and define a set of edges between nodes which represents the connection between brain regions. Each of these nodes contain millions of neurons which are assumed to serve a specific (cognitive) function. Here, network nodes are thus defined at the macroscopic scale.

2.2.3 Brain connectivity[30]

Recently there has been an increased interest in augmenting human brain mapping analysis with connectivity studies. This studies seek to describe how brain regions interact and how these interactions depend on experimental conditions and behavioural measures. Researchers seek to understand three types of connectivity as described below:

Structural connectivity

Structural connectivity describes anatomical connections which link a set of neural elements. These connections generally refer to white matter projections linking cortical and subcortical regions. In human neuroimaging studies, structural brain connectivity is commonly measured as a set of undirected links, since the directionality of projections currently cannot be discerned.

The structural connectivity has a relationship to functional and effective connectivity. the status of structural connectivity is an enhancing constraints or prior beliefs about effective connectivity. Effective connectivity depends on structural connectivity, but structural connectivity is neither a sufficient nor a complete description of connectivity. Structural connectivity changes over time, at microscopic and macroscopic levels[13].

Functional connectivity

In human brain, the integration of segregated areas is difficult to assess. Functional connectivity is one approach to characterize these integration, which is usually inferred on the basis of correlations among measurements of neuronal activity.

Functional connectivity describes patterns of statistical dependence among neural elements and generally derived from time series observations.Time series data may be derived with a variety of techniques such electroencephalography (EEG), magnetoencephalography (MEG), and functional magnetic resonance imaging (fMRI). Functional connectivity is an approach to characterize integration which is usually inferred on the basis of correlations among measurements of neuronal activity. Functional connectivity defines statistical dependencies among remote neurophysiological events. However, the correlations can arise in a variety of different ways. For ex, in multiunit electrode recordings, correlations can result from stimulus-locked transients evoked by a common input. Effective connectivity is used to better understand integration within a distributed system [13].

Effective connectivity

Effective connectivity attempts to capture a network of directed causal effects between neural elements. Effective connectivity refers explicitly to the influence that one neural system applies over another, either at a synaptic or population level. Effective connectivity replicate the observed timing relationships between the recorded neurons. The effective network represents patterns of causal interactions between neurons, as computed with transfer entropy which is a measure of directed information flow. The operational difference between functional and effective connectivity is important because it determines the nature of the inference about functional integration and the sort of questions that can be addressed. This distinction has played an important role in imaging neuroscience[13].

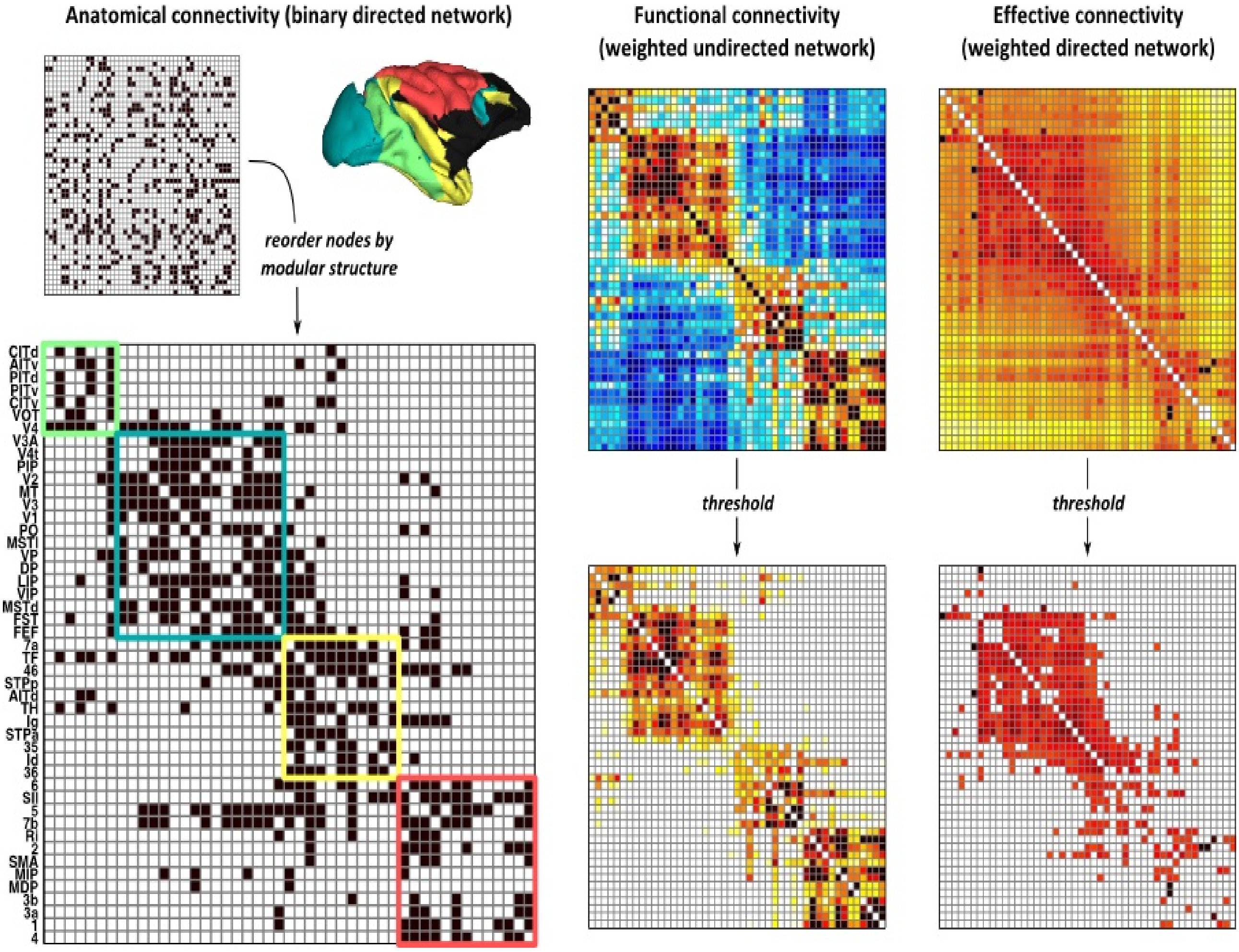

Figure 9: Illustrative anatomical, functional, and effective connectivity networks. This image is taken from the paper 'Complex network measures of brain connectivity: Uses and interpretations'

Figure 2 illustrates anatomical, functional, and effective connectivity networks. Large-scale connection pathways between cortical regions are represented by the anatomical network. Functional and effective connectivity networks were constructed from time series of brain dynamics simulated on this anatomical network. All networks are represented by their connectivity matrices where rows and columns in these matrices denote nodes, while matrix entries denote links. In connectivity matrices, the order of nodes has no effect on computation of network measures.

The nature of nodes

In brain network, nodes represents neural elements such as neuronal populations or brain regions. The brain regions with logical patterns of anatomical or functional connections is represented by nodes in the network. In individual brain network, the nature of nodes and links is determined by the combinations of brain mapping methods, anatomical parcellation schemes, and measures of connectivity.

The nature of links

Links are differentiated on the basis of type of connectivity such as anatomical, functional or effective and specific time scale features of connectivity. In addition to these factors, links are also differentiated on the basis of their weight and directionality. Presence or absence of directionality is an important factor to differentiate links, Thus, directed links may represents anatomical and effective connections conceptually.

Weighted links contains information about a connection strength and binary links denotes the presence or absence of connections between nodes. In anatomical networks, weights may represents the size, density, or coherence of anatomical tracts. In functional networks, weights represents magnitude of correlational interactions and weights in effective networks may represent respective magnitudes of correlational or causal interactions. Binary networks are in most cases simpler to characterize and have a more easily defined null model for statistical comparison, thus many recent studies discard link weights. Weight property of links are useful especially in filtering the influence of weak and potentially insignificant links. In functional or effective connectivity, weak and insignificant links may represents spurious connections. These links tend to vague the topology of strong and significant connection. Networks should correctly be characterized across a broad range of thresholds and threshold values are generally arbitrarily determined. Prior to network analysis, all self-connections or negative connections must be removed from network. Negative connections determines a functional anti-correlations between nodes.

2.3 Brain network measures [30]

An network measures play an important role to characterize and describe one or various aspects of functional integration and segregation, patterns of local anatomical circuitry, quantify importance of individual brain regions, and test resilience of networks to insult. Network measures are represented in different ways. Thus, the network measures of individual network elements such as nodes and links represents connectivity structures associated with these elements and follow the way in which these network elements embeds in the network. A distribution is comprises by measurement values of all individual elements in a network, which is characterized by its mean. These distribution provides a more global description of the network. For a nonhomogeneous distribution, other network feature such as distribution shape may be more important. In addition to these different parameters, network measures also have binary and weighted, directed and undirected variants. Figure 2 illustrates some basic concepts fundamental to these measures.

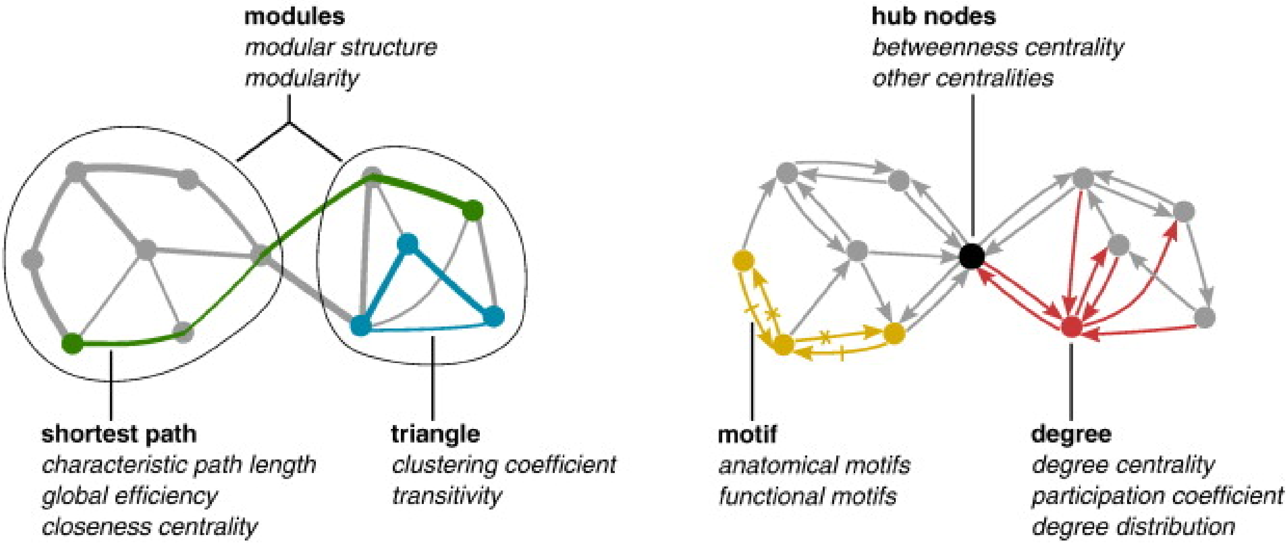

Figure 10: Measures of network topology. This image is taken from the paper 'Complex network measures of brain connectivity: Uses and interpretations

2.3.1 Measures of functional segregation [30]

clustering coefficient

The measures of segregation primarily measure the presence of densely interconnected groups of brain regions which is known as clusters or modules within the network. Functional segregation in the brain is defined as the ability for specialized processing to occur within densely interconnected groups of brain regions. In anatomical and functional networks, measures of segregation have a straightforward interpretations. In anatomical networks, the presence of clusters suggests the potential for functional segregation in these network, and in the functional network, presence of clusters denotes an statistical dependencies of segregated neural processing. The number of triangles in the network represents a simple measures of segregation where a high number of triangles implies segregation. Clustering coefficient is defined as the fraction of triangles around an individual node. The fraction of triangles around an individual node is equivalent to the fraction of the node's neighbors that are also neighbors of each other. The mean clustering coefficient for the network reflects the prevalence of clustered connectivity around individual nodes, on average. Mean clustering coefficient is normalized individually for each node. Thus, it may be disproportionately influenced by nodes with a low degree.

$$C = \frac{1}{n} \sum_{i \in N} C_i$$

where C_i is the clustering coefficient of node i, N is the set of all nodes in the network, and n is the number of nodes.

$$C_i = \frac{2t_i}{k_i(k_i - 1)}$$

Transitivity

Transitivity is defined as a classical variant of the clustering coefficient and it is normalized collectively. Both the network measures such as transitivity and clustering coefficient have been generalized for weighted and directed networks. Transitivity of network is not defined for individual nodes. Transitivity of network is as follows:

$$T = \frac{\sum_{i \in N} 2t_i}{\sum_{i \in N} k_i(k_i - 1)}$$

The network measures of segregation describes the presence of densely interconnected groups of regions and also find the exact size and composition of these individual groups which is known as the network's modular structure. It is revealed by subdividing the network into groups of nodes in such a way that there is a maximally possible number of within group links and a minimally possible number of between-group links. A modularity is a statistic which assess the degree to which the network may be subdivided into such non-overlapping groups. Unlike most of the other network measures, the optimization algorithm is used to estimate the optimal modular structure for a given network rather than computing exactly. Optimization algorithm generally sacrifices some degree of accuracy for computational speed.
Modularity of network is:

$$Q = \sum_{u \in M} [e_{uu} - (\sum_{v \in M} e_{uv})^2]$$

where e_{uv} is the proportion of all links that connect nodes in module u with nodes in module v, and the network is fully subdivided into a set of nonoverlapping modules M.

2.3.2 Measures of functional integration [30]

Characteristic path length

The ability to combine specialized information from distributed brain regions is a functional integration. This concept is characterized by network measures of integration by estimating the ease with which brain regions communicate. Measures of integration between different brain regions are commonly based on concept of a path. Paths are sequences of distinct nodes and links and in anatomical network, path represents routes of flow of information between two brain regions or nodes. The potential of functional integration between pair of nodes is represented by paths and strong potential of integration between brain regions are implied by shorter paths. Functional connectivity data already contains such information for all pairs of brain regions. In functional network, paths represents sequences of statistical associations. The most commonly used network measure of functional integration is average shortest path length between all pairs of brain region and it is known as a characteristic path length of the network.

Characteristic path length of the network is as follows:

$$L = \frac{1}{n} \sum_{i \in N} L_i$$

where L_i is the average distance between node i and all other nodes and N is the set of all nodes in the network, and n is the number of nodes.

$$L_i = \frac{\sum_{j \in N j \neq i} d_{ij}}{n-1}$$

Efficiency

Another related measure for functional integration is global efficiency which is the average inverse shortest path length. Characteristic path length is computed on a connected network, on the other hand, global efficiency may be knowingly computed on a disconnected network because paths between disconnected nodes are defined to have infinite length, and likewise zero efficiency. Global efficiency is influenced by short paths and the characteristic path length is basically influenced by a long paths. Binary path length is defined as a number of links in the path and the total sum of individual link lengths is a weighted path length. As large weights represents a strong associations and close proximity, link lengths are inversely related to link weights. Connection length do not represents spatial or metric distance and its typically dimensionless. Structural and effective network share a high global efficiency. Functional networks have lower global efficiency and weaker connections between the modules.

Global efficiency is as follows:

$$E = \frac{1}{n} \sum_{i \in N} E_i$$

where E_i is the efficiency of node i. Representation of E_i is as follows:

$$E_i = \frac{\sum_{j \in N j \neq i} d_{i_j^-} 1}{n-1}$$

2.3.3 Measures of centrality

Degree

Degree is an basic and important measure of a network. Number of links connected to that node is called a degree of a node, which is also equal to the number of neighbors of the node. The importance of a nodes in the network can be identified by the values of the degree. Degree distribution is comprises by degrees of all nodes in the network, which is an important marker of network development and resilience. An measure of density or the total 'wiring cost' of the network is commonly determined by the mean network degree. The weighted variant of the degree which is also sometimes termed as the strength is defined as the sum of all neighboring link weights. The basic network characteristics such as the number of nodes, links, and the degree distribution influences the values of many network measures. The degree is one of the most common measure of centrality. In functional integration, important brain regions often interact with many other regions and it plays a key role in network resilience to insult. Centrality measures of nodes diversely assess importance of individual nodes. The network measures motif and resilience sometimes used to detect central brain regions.

Mathematical definition for degree of node i in binary and undirected network is as follows:

$$K_i = \sum_{j \in N} a_{ij}$$

Where N is the set of all nodes in the network, and n is the number of nodes. (i, j) is a link between nodes i and j, where (i, j $\in N$).

a_{ij} is the connection status between i and j:

$a_{ij} = 1$ when link (i, j) exists, when i and j are neighbors; $a_{ij} = 0$ otherwise

($a_{ii} = 0$ for all i).

In Weighted and directed network, weighted degree of node is as follows:

$$K_i^w = \sum_{j \in N} w_{ij}$$

Directed out-degree of i is,

$$K_i^{out} = \sum_{j \in N} a_{ij}$$

Directed in-degree of i is,

$$K_i^{in} = \sum_{j \in N} a_{ji}$$

The connection weights w_{ij} is associated with the links (i,j). Therefore, we assume that weights are normalized, such that $0 \leq w_{ij} \leq 1$ for all i and j.

Network motif

In directed network, local patterns of connectivity are particularly diverse. For instance, anatomical triangles may consist of feedforward loops, feedback loops, and bidirectional loops, with different frequencies of individual loops likely having specific functional implications. These patterns of local connectivity are known as anatomical motifs. The frequency of occurrence of motif determines the significance of a motif in the network and it is usually normalized as the motif z-score. Motif fingerprint of that node is the frequency of occurrence of different motifs around an individual node and it reflect the functional role of the corresponding brain regions. The characteristic motif profile of the network is the frequency of occurrence of different motif in the whole network.

Betweenness centrality

Central nodes act as a central control of information flow among the nodes and many measures of centrality are based on idea of that central nodes participate in many short paths within a network. For instance, the inverse of the average shortest path length from one node to all other nodes in the network is called closeness centrality. The fraction of all shortest paths in the network that passes through a given node is called as a Betweenness centrality which is a sensitive and centrality related network measures. A high betweenness centrality is associated with the bridging nodes that connect disparate parts of the network. The notion of betweenness centrality associated with links is used to detect important anatomical and functional connections.

In anatomical and functional network, the measures of centrality may have different interpretations. For example, anatomically central nodes enable functional links between anatomically unconnected regions and often facilitate integration. Such links reduce the sensitivity of centrality measures in functional network and makes central nodes less noticeable. In most cases, weighted and directed variants of centrality measures are based on weighted and directed variants of degree and path length. Mathematical representation of betweenness centrality is as follows:

$$b_i = \frac{1}{(n-1)(n-2)} \sum_{h,j \in N h \neq j, h \neq i, j \neq i} \frac{p_{h_j}(i)}{p_{hj}}$$

where $_{hj}$ is the number of shortest paths between h and j, and $_{hj}(i)$ is the number of shortest paths between h and j that pass through i.

3 Method

In this section, the details of experiments, dataset for experiments, classification algorithms and methods that are used in single trial classification is described.

3.1 Machine Learning Techniques

3.1.1 Supervised Learning

Supervised learning is a type of Machine Learning which trains a model based on labeled input data. It is a technique for obtaining the input-output relationship information of a system through model training based on a given set of paired input-output labeled training samples which is also called as supervised data. Supervised machine learning is also referred to as a learning from teacher or learning from a labeled data. Supervised learning is also referred to as a inductive machine learning, which is a learning from observations. The goal of the supervised learning is to build an artificial system that can learn the mapping between the input and the output or that can learn from a given labeled data and it can predict the output of the system given new data inputs. The learning mapping leads to classification of the input data if the output takes a finite set of discrete values which indicates the class labels of the input. In supervised learning, supervised or labelled information is needed, while in unsupervised learning, labels are not given in training data while training a model. In semi-supervised learning, the algorithm uses both supervised and unsupervised training data

With supervised learning, machines can learn the human behaviour or object behaviour for certain tasks. The trained model or learned knowledge will then be used by machines or model to perform a similar actions on these tasks. Supervised learning allows machines to perform some input-output mappings much faster and more persistent than the human. So, machines equipped with a good supervised learner can perform certain tasks with more accuracy than humans. However, for many complicated tasks, existing Supervised Learning algorithms still cannot match human's learning ability on many complicated tasks because of limited hardware, software, and algorithm designs [25],

3.1.2 Evaluation metrics for data classification[17]

To achieve the optimal classifier during the classification algorithm and for evaluating the effectiveness of classifiers, the evaluation metric plays a critical role. Thus, the selections of proper evaluation metric is an important factor for selecting the optimal classifier. The evaluation metric can be described as the measurement tool that measures the performance of classifier. Accuracy is a common metrics used by many generative classifiers as a measure to differentiate the optimal solution during the classification algorithm. However, the accuracy has various weaknesses such as less distinctiveness, differentiability, informativeness and bias to majority class data. The evaluation metric can be classified into three types, which are threshold, probability and ranking metric which evaluates the classifiers with different goals. All these types of metrics are scalar group method where the complete performance is presented using a single score value. So it is easier to do the comparison and analysis. In practice, The most common metrics used by researchers to measure the performance of classifiers is the threshold and ranking metrics.

Threshold Types of Discriminator Metrics

The confusion metrics is used in binary classification problem for discriminating evaluation of the best optimal solution during the classification training. The confusion metrics is as shown in below figure.

	Actual Positive Class	**Actual Negative Class**
Predicted Positive Class	True positive (tp)	False negative (fn)
Predicted Negative Class	False positive (fp)	True negative (tn)

The row of the table represents the predicted class, while the column represents the actual class. In this confusion metrics, the number of positive and negative instances that are correctly classified is denoted by tp and tn. On the other hand, fp and fn denotes the number of misclassified negative and positive instances respectively.

Area under the ROC Curve (AUC)

AUC is one of the popular ranking type metrics which is used to construct an optimized learning model and also for comparing learning algorithms. Unlike the threshold and probability metrics, the AUC value reflects the overall ranking performance of a classifier. For evaluating the classifier performance and discriminating an optimal solution during the classification training, the AUC was proven theoretically and empirically better than the accuracy metric.

3.2 Experimental Data

In this project we used data that was used in a previous publication[21]. EEG data was collected from 54 schizo patients and 54 healthy controls. EEG were recorded using a 64-channel ANT digital EEG measure station (ANT, The Netherlands). Using the international 10/10 system, Ag-AgCl electrodes were arranged in an electrode cap. At a sampling frequency of 256 Hz, signals were digitised and it is stored for offline analysis. During EEG recording, participant performed a oddball task. In 'oddball task', subject hears a repeated and regular presentation of tones of perticular frequency, but the frequency of tones instead be a slightly different at some amount of time. Thus, P300 consists of 't' (target), 'd' (distractor) and 's' (standard) tones. The target and the distractor tones differ from the standard tone(1000 Hz) by a higher (1500 Hz) and a lower frequency (500 Hz) respectively. All tones have a duration of 100 ms and a loudness of 70 dB. These stimuli are presented pseudorandomly, with a distribution of 80% standard, 10% target and 10% distractor and an inter-stimulus interval randomised between 1 and 1.5 seconds. In total 400 tones are provided per test, with a total test time of 540 seconds. The ERP features can be amplitude and latencies of a set of peaks for the signals. ERP features can be considered as an important measures to use in training ML models for classification of target and distractor trials. In this project, ERP features and brain connectivity features are extracted from the data and these features are used for training a ML models for the classification of distractor and target trials.

Data is stored in a folder which is set up for each test set up. There are 54 folders for normal subject and 54 folders for schizophrenia subject. Each folder consists of matlab file which has same name as folder name, For ex. folder name for first normal subject is Norm01 which contains matlab file named as Norm01.mat. When this file is open in matlab, it creates a variable with same name as file name, For ex. When Norm01.mat will be opened in matlab, it will create a variable named as Norm01. These variable is matlab structure with 3 fields which corresponds to different stimuli used during the test. For p300 auditory, these will be called 'standard', 'target' and 'distractor'. Each of these fields contains matrix of dimension: 68 channels x 257 timepoints x n trails. The 68 channels correspond to 64 EEG electrodes + 2 extra channels for the Electrooculography + 1 channel for the Electrocardiography + 1 channel containing test stimuli. The order of the channels are: Fp1, Fpz, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, A1, T3, C3, Cz, C4, T4, A2, CP5, CP1, CP2, CP6, T5, P2, Pz, P4, T6, POz, O1, Oz, O2, AF7, AF3, AF4, AF8, F5, F1, F2, F6, FC3, FCz, FC4, C5, C1, C2, C6, CP3, CPz, CP4, P5, P1, P2, P6, PO5, PO3, PO4, PO6, FT7, FT8, TP7, TP8, PO7, PO8, HEOG, VEOG, EKG, Test. The 257 timepoints correspond to a time window from 200ms before stimulus until 800ms after stimulus acquired with a frequency of 256Hz. The number of trials vary in different subjects for 'distractor' and 'target' stimuli.

3.3 Feature detection

3.3.1 ERP Features

For ERP features, amplitudes and latencies of a set of peaks in channels Fz, Cz and Pz is detected. A peak is defined as a time point that has a higher amplitude than its neighboring time points. Multiple peaks are detected in an interval. Peaks and detection intervals for peaks is as follows:

Paradigm	Event	Peak	Detection Interval (ms after stimulus)
Auditory P300	target	N100	[50, 160]
		P200	[100, 250]
		N200	[150, 310]
		P300	[260, 450]
	distractor	N100	[50, 150]
		P200	[100, 260]
		N200	[140, 300]
		P300	[260, 450]

Figure 11: Detection interval for peaks [21]

As shown in table, negative peaks are detected as N100 and N200 and positive peaks are detected as P200 and P300 in the respective detection interval of target and distractor event. The complete feature set consists of amplitude and latency of each of these peaks in channels Fz, Cz, and Pz which yields 6 features per peaks in 1 interval. For four different detection intervals, 24 features are extracted for each target and distracted trials.

3.3.2 Connectivity Features

To obtain connectivity features, first connectivity matrices are extracted from each single trial of target and distractor and for first 64 channels. The conneectivity matrices represents the correlation of the EEG channels within a trials. These connectivity matrices can then be used to extract connectivity features for each trial using brain connectivity toolbox. These connectivity features will then be used for single trial classification. Following brain connectivity features are extracted:

- **Betweenness bin**

  Betweenness centrality is a measure of centrality which is defined as the fraction of all shortest paths in the network that pass through a given node.

- **Characteristic path length**

  Characteristic path length of the network is the average shortest path length between all pairs of nodes in the network.

- **Clustering coefficient**

  Clustering coefficient is The fraction of triangles around an individual node in the network.

- **Degree**

  Degree of a node is the number of links connected to that node or number of neighbors of that nodes.

- **Efficiency**

  Efficiency in the network is the average inverse shortest path length.

- **Modularity**

  The network measures of segregation describes the presence of densely interconnected groups of regions and also find the exact size and composition of these individual groups. A modularity is a statistic which assess the degree to which the network may be subdivided into such non-overlapping groups.

- **Transitivity**

  The classical variant of the clustering coefficient is known as transitivity.

3.4 Dataset preparation for ML model

Features extracted in Matlab are managed by creating a Pandas DataFrame for ERP and connectivity feature values for each target and distractor trials. There are four mat files of features are created in MATLAB: for single control, group of 10 controls, for single schizo and group of 10 schizo subject. These files loaded into a python file using loadmat python library. Distractor dataframe is created using distractor features and dataset is labelled as 0, then target dataframe is created using target features and dataset is labelled as 1. Distractor dataframe and target dataframe are concatenated into single dataframe for applying machine learning algorithms. This process is applied for all the subject types such as 1 normal, 1 schizo, 10 group of normal, and 10 group of schizo subject. On the resulting dataframes nested cross validation is applied to make sure that feature selection and feature preprocessing processes are based only on training portion of data and preventing a leakage from the validation set and test set.

3.5 Nested cross validation

For a particular problem, a scientists who builds a classification model has to select the best algorithm. There are many classification algorithm such as k-nearest neighbors, SVM, naive bayes, neural network, decision tree, random forest classifier and so on. When tackling a certain problems, there are the theoretical and experimental reasons to prefer a particular algorithm over another. While solving particular problems using ML, we can not predict on priori based on current understanding of ML, which algorithm will perform better or solve the problem with good accuracy. Therefore to find out which algorithm is giving good performance, one should benefit from trying many competing ML algorithms. However, most of the ML algorithms have one or more hyperparameters that must be set to certain optimal values to increase the model accuracy. For ex. The k-nearest algorithm is having one hyperparameter which is a k, and k refers to number of neighbors used while training a model. On the other hand, random forest algorithm is having at least 2 hyperparameters which is the number of trees to be constructed and the number of features considered at each split. Selecting an algorithm for specific problem and its hyperparameter tuning are dependent processes. Algorithm performs very well with a particular set of hyperparameters, on the other hand it may perform worse than other algorithms with different set of hyperparameters. Therefore it is important to chose both the algorithms and its hperparameters in such a way that it will maximize its expected performance on future data.

We can usually split the data repeatedly rather than only once into a training, and test set to determine the test accuracy, sensitivity, or specificity. The data can be randomly split into k disjoint sets of approximately equal size which is called as folds. Each fold is used once as a test set while all the other folds are combined and then can served as the training set. This process is called as a k-fold cross-validation. Cross-validation is the method of choice for assessing the classifiers performance on previously unobserved data. The most commonly used cross-validation schemes are leave-one-out cross-validation in which k is equals to the number of number of data samples and 10-fold cross-validation in which k is equal to 10. For most of the datasets, 10-fold cross-validation is used with regard to bias and variance where bias represents the expected difference of the classifier's prediction as compared to the true class-membership and variance represent the variability of the classifier's prediction for one data samples[20].

Choosing an appropriate model and optimising the hyper-parameters are most often performed by minimising a cross-validation estimate of generalisation performance. k-fold cross-validation is the most basic form of cross validation which partitions the available data into k disjoint chunks of approximately equal size. In each iteration, a training set is constructed from a different combinations of k-1 subsets, and remaining subset is used as the test set. After constructing a training and test set, a model is fit to the training set and its performance is evaluated on test set. For obtaining a estimate of the generalization performance of a model fit for all the available data, we can take the average of the performance metric on the test set in each iteration. For selecting a best algorithm and tuning the hyper-parameters through cross-validation, we can use a method called as nested cross-validation, which is also known as double cross-validation[20].

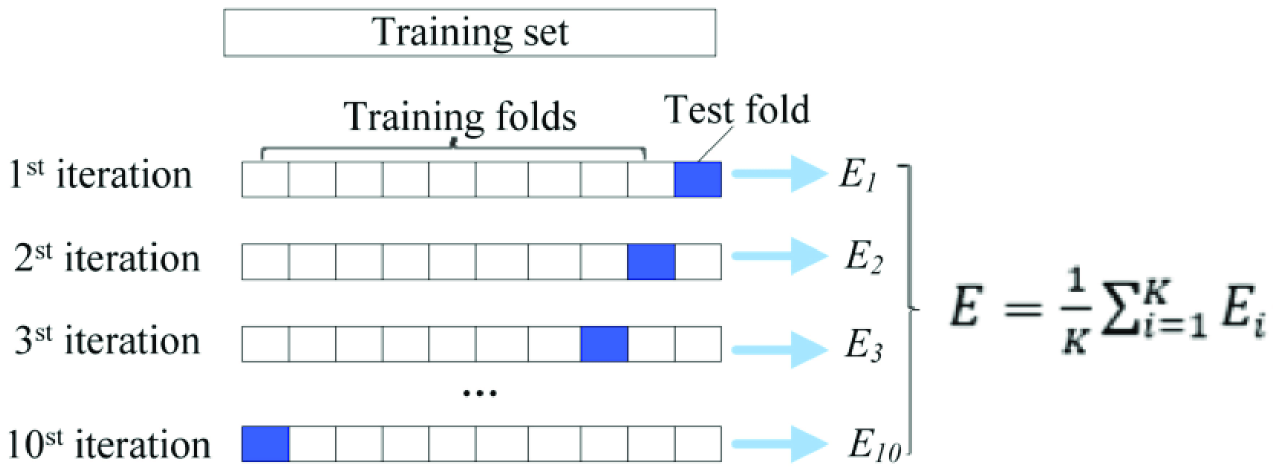

Figure 12: Cross validation. This image is retrieved from the article 'RFAmyloid: A Web Server for Predicting Amyloid Proteins' available via license: Creative Commons Attribution 4.0 International

We can use cross-validation in combination with feature selection and also for the selection of tuning parameters for the classifiers. The parameters of classifiers to be tuned are such as the penalization parameter in lasso-regularized logistic regression. In such case, to avoid optimistically biased performance estimates, we can use nested cross validation. The idea of nested cross validation is to start with regular k-fold cross-validation which is called as outer loop to assess the final classifier performance. In cross validation, each fold is used as once as the test set and all the other folds are then used as the training set. We will then split each training set randomly again and which we call as inner cross-validation loop. We can use this inner-cross validation loop for feature selection or for tuning parameter selection. Thus, we can use the training and test sets from inner loop to try out different feature subsets or classifier tuning parameters. The best performing classifier from this level is then applied to test set from the outer loop. the process of nested cross validation is as shown in below figure 13[20].

In nested cross validation, we have a outer loop and inner loop. An outer cross-validation procedure is performed to provide the performance estimate used to select the optimal model. Using each outer cross-validation training fold, we can perform a tuning of hyperparameters, and/or preprocessing and feature selector of the model independently to minimize an inner cross-validation estimate of generalisation performance. The outer cross-validation is then essentially estimate the performance of a method for fitting a model. In outer loop, we divide the data into training and test set, and train the model on training data. In inner loop, we divide the training data into training and validation set, and we will then use the validation set to perform the parameter optimization. This eliminates bias introduced by the cross-validation. This is because, the test data in each iteration of the outer cross-validation has not been used to optimise the hyperparameters of the model, which provides the more reliable criteria for choosing the best model.

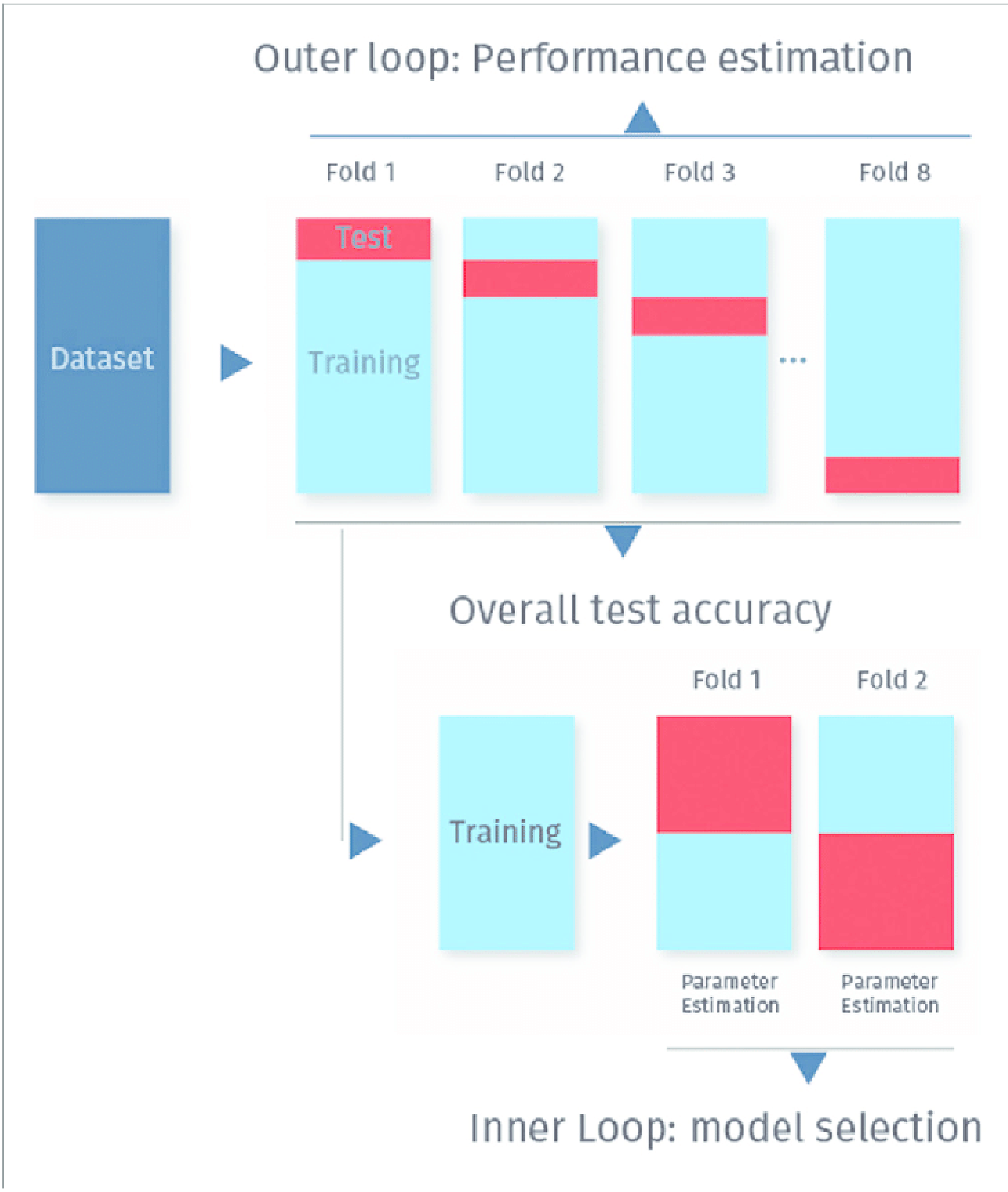

Figure 13: Nested cross validation. Retrieved from the article 'Promises, Pitfalls, and Basic Guidelines for Applying Machine Learning Classifiers to Psychiatric Imaging Data, with Autism as an Example'

3.5.1 Pipeline and GridSearchCV

In a machine learning workflow, we need to implement following key steps.

- **feature preprocessing** These is the most important step in machine learning, which involves encoding categorical features, scaling numerical features, transforming text data, etc.

- **feature selection** In feature selection process, We can choose which feature to include in the model. It is a process of selecting a subset of relevant features for construction of model. In machine learning, feature selection is also known as variable selection, attribute selection, or variable subset selection.

- **Model selection** It is process of choosing which machine learning estimator should be used. It is a task of selecting a statistical model from a given set of competing models on a given data for a particular problem.

- **Hyperparameter tuning** It is a procedure which determines the optimum hyperparameter values to use for each estimator.

To perform these processing in an accurate, efficient, and reproducible manner is an challenging tasks. It is important to ensure that, during cross-validation , feature selection and feature preprocessing processes are based only on training portion of data and prevent a leakage from the validation set and test set. Leakage from validation set or test set during feature selection or preprocessing could lead to the bias for our results. We can use a scikit-learn to simultaneously perform a feature selection, feature preprocessing, model selection, and hyperparameter tuning,

Pipeline framework

We need to first set a pipeline object that will perform a following tasks:

- **StandardScalar** StandardScalar is used to standardize the data.

- **feature selection** For performing feature selection, we will use 'SelectKBest' and mutual information metric. It will select the k best features from data. Here, k is the hyperparameter that will tune during the fitting process.

- **Model selection** For model selection, we are trying a machine learning estimators such as LogisticRegression, DecisionTree, RandomForestClassifier, support vector machine, and KNeighborsClassifier.

To instantiate a pipeline object, we have used the value of 'k' in SelectKBest as 5. The syntax for creating a pipeline is as follows:

```
pipe = Pipeline([('scaler', StandardScaler()),
('selector', SelectKBest(mutual_info_classif, k=5)),
('classifier', LogisticRegression())])
```

Search Space

We can then define the space of hyperparameters and estimators which you want to search through. We can use a dictionary to do this and will use a underscore notation (_) to refer to the hyperparameters of different steps in our pipeline. We can try different hyperparameters for each estimators and different values for k for the feature selector SelectKBest. The estimators used in this experiments and its respective hyper-parameter values is as shown in following search space. The estimators used in this experiments are LogisticRegression, SVM, KNeighborsClassifier, DecisionTreeClassifier, and RandomForestClassifier with hyperparameters as C, C, n_neighbors, max_depth, and max_depth respectively. The hyperparameter C of logistic regression represents the strength of the regularization. The higher value of C correspond to less regularization. The C parameter in SVM is a regularization parameter which controls how much you want to punish your model for each misclassified point. The hyperparameter n_neighbors specifies the number of neighbors used in KNearestNeighbor classification algorithm. It specifies how many nearest points the algorithm should look for assigning a class label to new data point. The hyperparameter for decision tree is max_depth which determines the maximum depth of a tree. We can use this hyperparameter for decision tree to prevent over-fitting as using max_depth parameter, we can regularize the tree or limit the way it grows to prevent the over-fitting. The max_depth hyperparameter in random forest specifies the maximum depth of each tree. the default value for max_depth parameter of random forest is None, which means each tree will expand until every leaf is pure and pure leaf specifies a node where all of the data on the leaf comes from the same class.

3.5.2 Feature preprocessing

Standardisation step is performed on a feature values before classification started. Standardizing dataset involves rescaling the distribution of values so that the mean of observed values is 0 and standard deviation is 1. It is a technique where the values are centered around the mean with a unit standard deviation. In the standardisation step, for each instance the feature mean was subtracted from the individual subject's feature value and this result was divided by the standard deviation of that feature which is called a z-score. Features are transformed into their z-scores to ensure that all features have equal weight in training. Classifiers gets biased(ref) if features do not have equal weights.

3.5.3 Feature selection

Feature selection is an important technique in machine learning where we choose those features in data that contributes most to the target variable. With feature selection techniques we choose the best predictors for predicting the target variables. The classes in the sklearn.feature_selection module can be used for feature selection or dimensionality reduction on sample sets. Feature selection is a technique of removing features in the training data so as are statistically uncorrelated with the class labels. Feature selection technique will also reduces the set of features which are used in classification and selects those features which has highest correlation with target variable. Feature selection is a term which is also called as the variable subset selection, variable selection, attribute selection, or feature reduction. It also helps to improve efficiency and to reduce the dimensionality and accuracy of machine learning classification algorithm. This techniques identifies the fields that are most important in predicting a certain outcome. Thus, to reduce the amount of training data needed to obtain the desire level of performance and to improve the classification accuracy, feature selection technique is applied. Feature selection is frequently used as a preprocessing step to machine learning based on statistical theory. Moreover, feature selection is a process of choosing a subset of original features so that the feature space is optimally reduced according to a certain evaluation criteria.

One of the advantage of feature selection is that we will use fewer predictors in feature selection and using fewer predictors is less expensive and computing time is also improving as we are using fewer predictors. With feature selection, the predictors which contributes less in prediction can be skipped from the dataset which results in quicker, easy to understand, and more efficient model that makes the use of fewer predictors. Feature selection determines the relevant features by their relations with the corresponding class labels and discards irrelevant and redundant features. To choose the relevant features, unsupervised feature selection explores data variance. For supervised feature selection, two methods namely, wrapper and filter algorithm is used for the existing feature subset selection. The main difference between the two methods is that filter algorithms select the feature subset before the application of any classification. Filter algorithm uses statistical properties or probabilistic characteristics of features in filter to remove the less important feature from the subset based on their comparative computational efficiency. Filter methods define the relevant feature without prior classification of data and are independent of the learning machines. The wrapper method approach using machine learning and selecting the feature subset according to the accuracy on the training data and testing the classification using the test data. The wrapper method is more time consuming in compare with the filter method[10].

Feature selection using ANOVA[14]

To improve the machine learning classifier accuracy scores or to boost their performance on a very high-dimensional datasets, we can use the classes in the sklearn.feature_selection module for feature selection or dimensionality reduction on a sample sets. ANOVA is a feature selection technique which helps to select the best features from the data.

Variance is the measurement of the spread between numbers in a variable. Variance measures how far a number is from the mean and every number in a variable. The variance of a feature determines how much the variable is impacting the response variable. If the variance is low, it implies there is no impact of this feature on response and vice-versa.

Analysis of Variance is a statistical method which is used to check the means of two or more groups that are significantly different from each other. For the analysis of variance, a probability distribution generally used. It assumes the hypothesis as: two variances are equal and two variances are not equal.

To increase the score of model, we need a dataset that has high variance. So it will be good if we select a feature that has higher variance and which has higher impact on target. We can achieve this using ANOVA filter. SelectKBest selects features according to K highest score. We can use f_classif function in SelectKBest for using ANOVA filter.

3.5.4 Classification

In this section we describe the algorithms that we choose to use for our classification purposes namely, Support Vector Machines with linear kernel, K-Nearest Neighbors and Multinomial Naive Bayes are briefly explained.

Decision tree

A well known machine learning classifier is decision tree classifier which generates a tree structure in which each node represents a split of dataset based on feature's value. This tree is traversed using all feature values of an instance and final classification of that instance is the class at the end of path. The classification algorithm generates such a trees based on information gain calculation for individual features [21].

A decision tree is a flow-chart-like hierarchical tree structure which consists of decision nodes which corresponds to attributes, edges which corresponds to different possible attribute values and leaves including objects that belongs to same class or that are very similar [19]. A tree has three types of nodes: root node that has no incoming edges and zero or more outgoing edges, Internal nodes which has exactly one incoming edge and two or more outgoing edges and Leaf or terminal nodes which has exactly one incoming edge and no outgoing edges. Decision tree classify the example by sorting them down the tree from the root node to some leaf node which gives classification to the example. Each node in a tree acts as a test case for some attributes and each edge descending from that node corresponds to one of possible answers to the test case. This process is recursive and repeated for every subtree [6] . The pruning technique is implemented on tree when the tree is fully built. Pruning is a technique of cutting branches in the tree that add too little extra accuracy. Cutting these branches reduces the risk of overfitting and use of random features. A structure of decision tree is as shown in below figure.

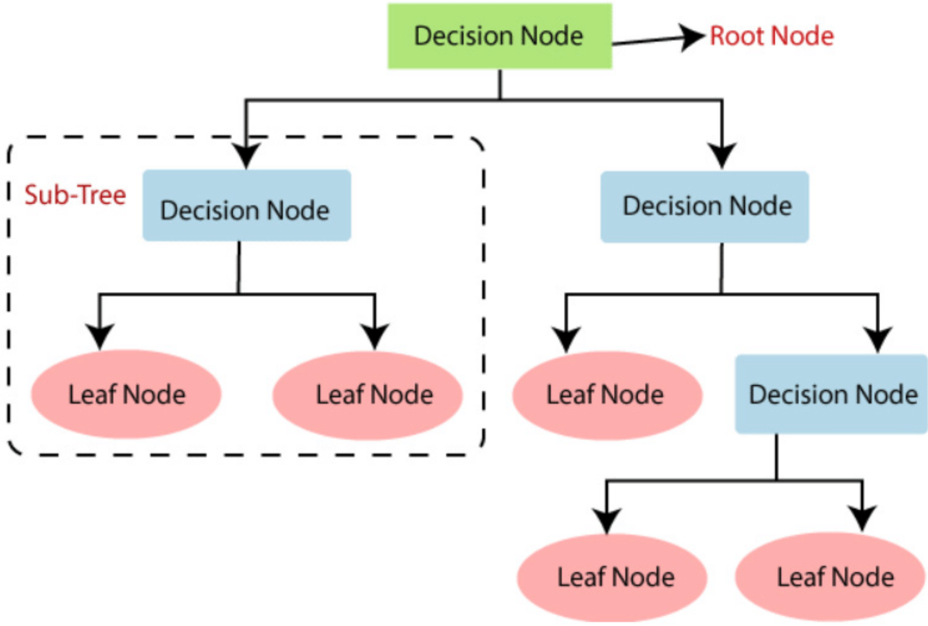

Figure 14: Structure of decision tree[16]

Random forest

A random forest fits a number of decision tree classifiers on different sub samples of the dataset and uses averaging of classifiers to improve the predictive accuracy and to control overfitting max_sample parameter is used to control the sub-sample size, if bootstrap=True (default), otherwise the whole dataset is used to build each tree. For each of individual trees from a collection of randomly generated decision trees, a subset of features is selected at random from the complete feature set and then tree is built. Training a decision tree is difficult as there is a large number of features. Random forest classifier resolves this problem of decision tree. The prediction is based on majority voting of all trees after a fixed number of tress has been built. Random responses are generated from useless trees in a collection and most of trees are useless, but on average these will cancel each other out, which result in only the relevant trees adding to the prediction(1). “randomForest” from sklearn is used for this classifier. A general random forest architecture is as shown in below figure.

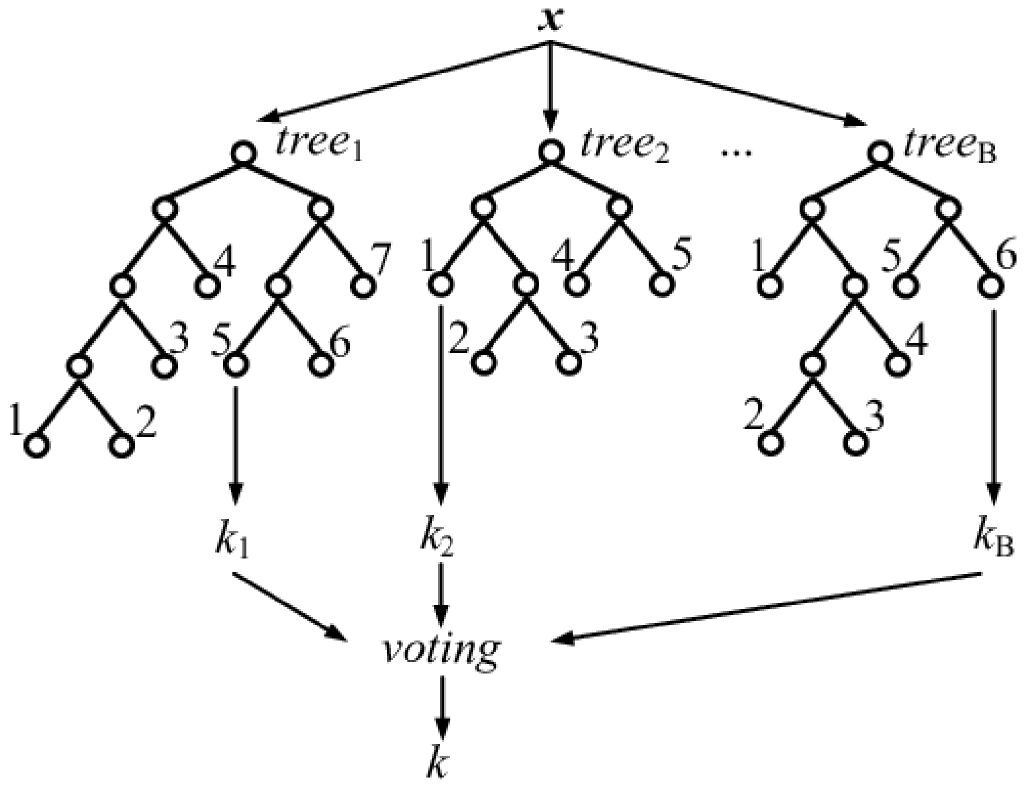

Figure 15: Random forest architecture[15]

Support Vector Machines(SVM)

A support vector machine is a supervised machine learning model that is able to generalise between two different classes. SVM checks for that hyperplane that is able to distinguish between the two classes. There can be many hyperplanes that can classify the point, but the objective is to find the hyperplane that has highest margin i.e. maximum distances between the two classes.[9]

In this experiment, a linear kernel is used in SVM algorithm. optimal hyper-parameter such as C and gamma can be found by just trying all combinations and see what parameters work best. The main idea behind it is to create a grid of hyper-parameters and just try all of their combinations. We have used GridSearchCV from the scikit library to determine C and gamma hyperparameters which ensure a good result for the classifier. C being the regularization parameter of the error term. A low C makes the decision surface smooth, while a high C aims at classifying all training examples correctly. While gamma defines how much influence a single training example has. The larger gamma is, the closer other examples must be to be affected.

SVM is as shown in below figure. SVM is a linear machine which operates in k-dimensional space created due to n-dimensional input data X into k-dimensional space using non-linear mapping which helps to isolate data normally by geometry and linear algebra. To find a linear classifier for any data points with known class labels, this can be done by identifying a separating hyper plane [28].

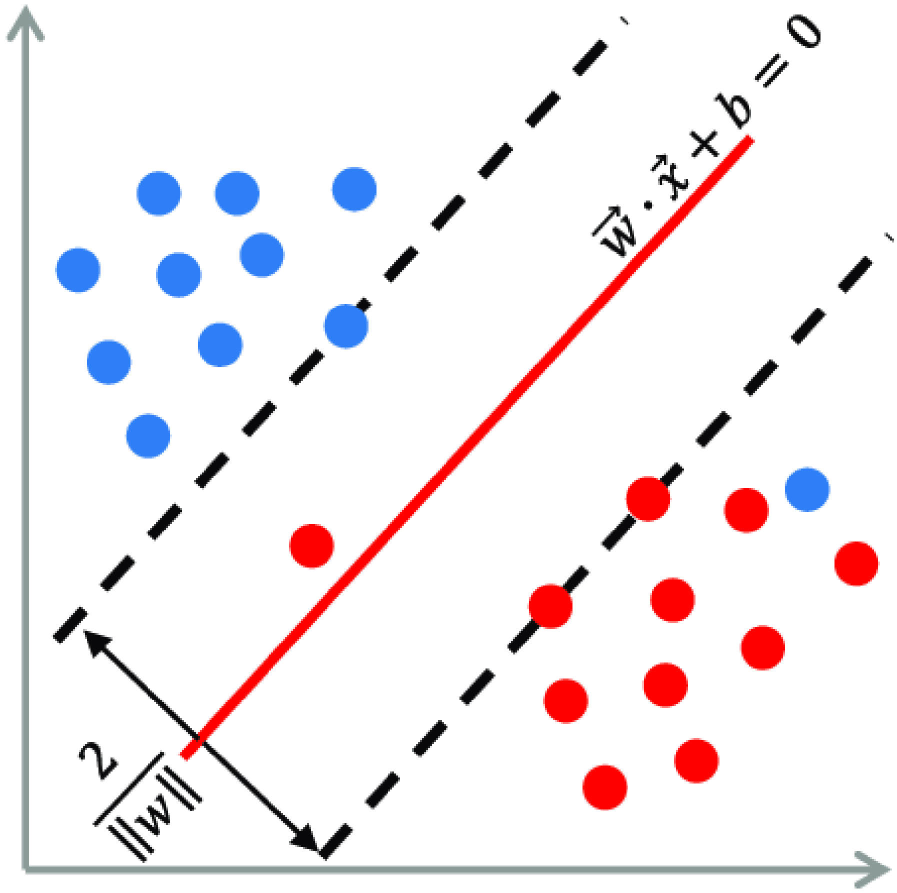

Figure 16: Support Vector Machine (SVM)[24]

K-Nearest Neighbors

The K-Nearest Neighbors algorithm performs quite complex classification tasks though its extremely easy to implement in its most basic form. KNN does not have any specialized training phase. it uses all of the data for training while classifying a new data point or instance. It is a non-parametric learning algorithm. This means that it does not assume anything about the underlying data. This is the most useful feature about the data because many of the real world data doesn't follow any theoretical assumption. As a example, we can say that assumptions such as linear-separability, uniform distribution, etc.[29]

- **Distance measurement** : We can calculate the Euclidean distance as the square root of the sum of the squared differences between a new point and an existing point across all input attributes. This is most widely used for distance measurement for the new data point from all the other data points.

- **Feature Scaling** : It is always a good practice to scale the features, before making any actual predictions, so that all of them can be uniformly evaluated. The gradient descent algorithm also converges faster with normalized features.

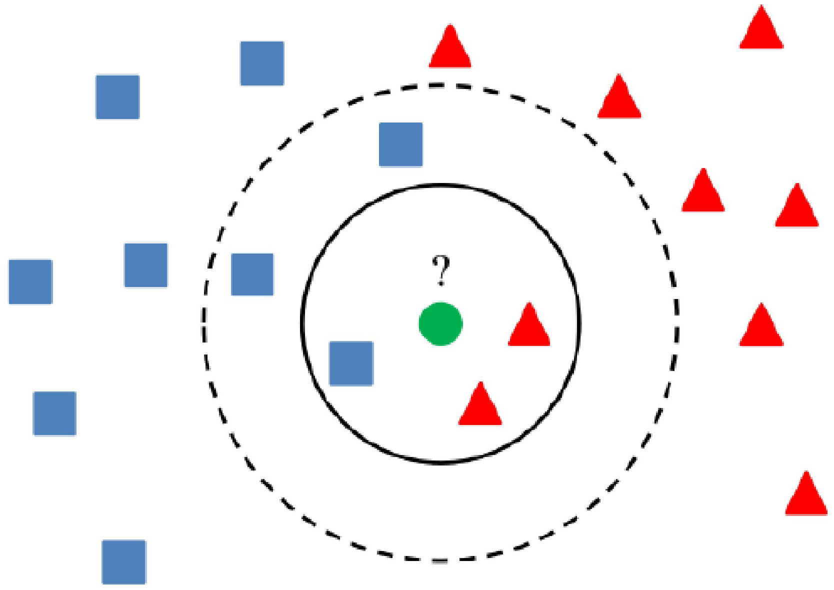

Figure 17: K-nearest neighbor algorithm [1]

K-nearest neighbor algorithm is as shown in above figure. In this figure, the green circle is the sample which is to be classified. Blue squares and red triangles illustrates samples from two different classes in the training set. If k is 3, then the 3 nearest neighbors will be considered to decide which class the analysed sample will be assigned. If K is 5, the 5 nearest neighbors will be considered.

Logistic Regression

In medical research, Logistic regression (LR) analysis has become an increasingly employed statistical tool which is used in a situation where a binary outcome is to be predicted from one or more independent or predicting variables. In research method, rather than focusing on when the event occurred, Logistic regression model is used when the focus is on whether or not an event occurred. It is particularly used for the models which involves disease state such as whether disease is present or it is not present, or for a decision making such as either 'yes' or 'no'. As LR provides binary classification results, it is widely used in studies in the health science. In many complex problems, there may be a situation where the predicted variable takes more than two variables, which is called as polychotomous or multinomial logistic regression. In order to fit a model to the data, certain assumptions are made. LR assumes relation between the logit of the outcome and the predictor values, but it does not assume the linear relationship between the dependent and independent variables. In LR, the dependent variables must be categorical, and the independent variables need not be interval, normally distributed, linearly related, and with equal variance within each group. LR can contain both categorical and continuous independent variable, however the power of analysis of data with LR will be increasing when the independent variables are normally distributed and do have a linear relationship with the dependent variables. For a specific problem, LR computes the probability that a case with a particular set of values for the independent variables is a member of modeled category. LR has many applications in the field such as physical sciences, economics, and political sciences. LR has been increasingly being applied in medical research. The use of LR in medical sciences includes a study of the factors that predict whether an improvement or no improvement will occur after an intervention, the presence or absence of a disease in relation to a variety of factors, to determine which of a range of potential predictors actually are important, to explore the effects of and relationships between multiple predictors, and to determine whether newly explored variables add to the predictive validity of already established models [3].

3.6 Feature Importance

In ML and statistics, feature selection is also known as variable selection, attribute selection or variable subset selection. It is the process of selecting a subset of relevant features, variables, or predictors for the use in model construction. In ML data may contain redundant or irrelevant features, thus it can be removed without causing much loss of information. So, we can use feature importance techniques for identifying those features and using important features in model training to get improving accuracy. In ML it is important to have a better accuracy but it is also equally important to have an interpretable model. Along with knowing how much accuracy our model is giving for classifying the instances, it is good to know why our model is giving this high/low accuracy and which features are most important in classifying instances. Identifying feature importance are helpful for improving model accuracy and early detection. With feature importance, we can get a better understanding of model's logic and with feature importance we can also focus on most important features of model which helps in improving model accuracy. With feature importance, we can also find the most significant features which has a higher impact on target variable and we can then remove the variables that are less significant for training the model[22].

We have hundreds or millions of features in data science and we want a way to create a model that only includes the most important features. The benefit of choosing the most important features is that: First, we make our model more simple which will be easy to interpret, Second, we can reduce the variance of model with which we can avoid overfitting of our model. And third, we can reduce the computational cost and time for training the model. Feature selection is the process of identifying the most relevant features. In a data science workflow, we often use random forest for feature selection because the random forest uses tree-based strategies and these strategies by random forest naturally ranks by how well they improve the purity of the node. This means decrease in impurity over all trees which is called gini impurity. Nodes with the greatest decrease in impurity happen at the start of the trees. On the other hand, nodes with the least decrease in impurity occur at the end of trees. Thus, we can create a subset of the most important features by pruning trees below a particular node. Feature importance assigns the score of input features based on their importance to predict the output. The score of input features will be more if they are more responsible for to predict the output. [2].

4 Results

We measured the performance of classifiers for ERP features and connectivity features for classifying single-trial classification of ERP data. For measuring the performances of classifiers, we calculated accuracy, sensitivity, and specificity using nested cross validation. Using nested cross validation, we have used simultaneous processing of feature selection, feature standardization, and model selection. The classifier which we used in this experiments are logistic regression, decision tree, random forest, k nearest neighbor, and SVM. We have first calculated accuracy, sensitivity, and specificity for all classifiers and we then obtained the mean of the result from all classifiers. From the experiments, it is observed that connectivity features are giving better accuracy than ERP features for a group of 54 normal and 54 schizophrenia subject types. For ERP features, the mean and standard deviation of accuracy is 0.5899 (0.0262) in normal subject and it is 0.5057 (0.0282) in schizophrenia subject. For connectivity features, the mean and standard deviation is 0.5920 (0.0263) in normal subject, and it is 0.5671 (0.0255) in schizophrenia subject. If we compare this results, connectivity features are giving more accuracy than ERP feature. It is also observed from the experiments that random forest classifier in general is giving better accuracy than other classifiers. The brief results for these group analysis with ERP and connectivity features are as shown in below table.

4.1 Performance with ERP features

4.1.1 Performances for group of 54 subject

Table 1: Result with ERP features for group of 54 subjects

Classifier	Performances	group of 54 normal subject	group of 54 schizophrenia subject
Logistic Regression	Accuracy (SD)	0.5948 (0.0238)	0.5153 (0.0171)
	sensitivity (SD)	0.5699 (0.0534)	0.4852 (0.0424)
	specificity (SD)	0.6187 (0.0453)	0.5450 (0.0290)
Random Forest Classifier	Accuracy (SD)	0.5936 (0.0283)	0.5105 (0.0240)
	sensitivity (SD)	0.5712 (0.0510)	0.4836 (0.0464)
	specificity (SD)	0.6151 (0.0514)	0.5371 (0.0614)
Decision tree Classifier	Accuracy (SD)	0.5877 (0.0315)	0.5049 (0.0224)
	sensitivity (SD)	0.5655 (0.0525)	0.4711 (0.0855)
	specificity (SD)	0.6091 (0.0489)	0.5382 (0.0479)
KNeighbors Classifier	Accuracy (SD)	0.5874 (0.0225)	0.5054 (0.0255)
	sensitivity (SD)	0.5549 (0.0449)	0.5124 (0.0379)
	specificity (SD)	0.6187 (0.0441)	0.4985 (0.0450)
SVM Classifier	Accuracy (SD)	0.5936 (0.0257)	0.5119 (0.0140)
	sensitivity (SD)	0.5730 (0.0432)	0.5028 (0.0364)
	specificity (SD)	0.6133 (0.0480)	0.5209 (0.0315)

4.1.2 mean and standard deviation of accuracy from all classifiers

Table 2: Mean and standard deviation ERP features		
Subject Types	Performances	mean (standard deviation)
For group of 54 normal subject	Accuracy	0.5899 (0.0262)
	sensitivity	0.5718 (0.0443)
	specificity	0.6073 (0.0403)
For group of 54 schizophrenia subject	Accuracy	0.5057 (0.0282)
	sensitivity	0.5113 (0.0398)
	specificity	0.5002 (0.0464)

Above table shows the mean and standard deviation from all classifiers using ERP features.

4.2 Performance with connectivity features

4.2.1 Performances for group of 54 subject

Table 3: Result with connectivity features for group of 54 subjects

Classifier	Performances	group of 54 normal subject	group of 54 schizophrenia subject
Logistic Regression	Accuracy (SD)	0.5945 (0.0282)	0.5713 (0.0245)
	sensitivity (SD)	0.5448 (0.0469)	0.5573 (0.0227)
	specificity (SD)	0.6422 (0.0345)	0.5852 (0.0372)
Random Forest Classifier	Accuracy (SD)	0.6074 (0.0253)	0.5643 (0.0221)
	sensitivity (SD)	0.5217 (0.0525)	0.5442 (0.0230)
	specificity (SD)	0.6897 (0.0428)	0.5841 (0.0405)
Decision tree Classifier	Accuracy (SD)	0.5936 (0.0345)	0.5809 (0.0210)
	sensitivity (SD)	0.5392 (0.0350)	0.5528 (0.0131)
	specificity (SD)	0.6458 (0.0511)	0.6087 (0.0382)
KNeighbors Classifier	Accuracy (SD)	0.5905 (0.0254)	0.5801 (0.0347)
	sensitivity (SD)	0.5643 (0.0480)	0.5544 (0.0313)
	specificity (SD)	0.6157 (0.0391)	0.6053 (0.0486)
SVM Classifier	Accuracy (SD)	0.5887 (0.0308)	0.5770 (0.0204)
	sensitivity (SD)	0.5342 (0.0559)	0.5556 (0.0250)
	specificity (SD)	0.6410 (0.0344)	0.5980 (0.0472)

4.2.2 mean of accuracy from all classifiers

Table 4: Mean and standard deviation for connectivity features		
Subject Types	Performances	Result
For group of 54 normal subject	Accuracy	0.5920 (0.0263)
	sensitivity	0.5216 (0.0574)
	specificity	0.6608 (0.0558)
For group of 54 schizophrenia subject	Accuracy	0.5671 (0.0255)
	sensitivity	0.5067 (0.064)
	specificity	0.6260 (0.057)

The experiment is also run for the different subject types such as for single normal subject, single schizophrenia subject, group of 10 normal subject, and group of 10 schizophrenia subject for both the feature sets. The performances from these subject types are summarized in the Appendix section of this thesis.

4.3 Feature Importance

In this experiment, feature importance are calculated using random forest algorithm for group analysis of 54 subject. Feature importances represents relevance of our features. In this thesis, these ranking does not use in seperate step of feature selection. It is measured to get feature importances in both feature sets to determine how much the variables are impacting in model permance. Below is the feature importances and its graph for ERP and connectivity features for respective groups of normal and schizophrenia subject.

4.3.1 Feature importance ERP Feature

Below values and graph shows feature importances for all 24 ERP features in normal and schizophrenia subject respectively. To name ERP features, channel name, 'max' keyword for maximum, 'amp' for amplitude, and a number for time interval such as 1 for first time interval, 2 for second time interval, and so on is used. Different time intervals are as shown in figure 11 for both target and distractor.

For ex. the feature name 'max_ampcz1' represents maximum amplitude of peak for channel cz for first time interval and 'latency_maxcz1' represents maximum latency of peak for channel cz for first time interval.

for a group of 54 normal subject

```
Feature: 0, Score: 0.04457
Feature: 1, Score: 0.04182
Feature: 2, Score: 0.04440
Feature: 3, Score: 0.04469
Feature: 4, Score: 0.03948
Feature: 5, Score: 0.04297
Feature: 6, Score: 0.04542
Feature: 7, Score: 0.04116
Feature: 8, Score: 0.04593
Feature: 9, Score: 0.04881
Feature: 10, Score: 0.05613
Feature: 11, Score: 0.05326
Feature: 12, Score: 0.03403
Feature: 13, Score: 0.03114
Feature: 14, Score: 0.03151
Feature: 15, Score: 0.03749
Feature: 16, Score: 0.03348
Feature: 17, Score: 0.03534
Feature: 18, Score: 0.04103
Feature: 19, Score: 0.03538
Feature: 20, Score: 0.03652
Feature: 21, Score: 0.05162
Feature: 22, Score: 0.04224
Feature: 23, Score: 0.04158
```

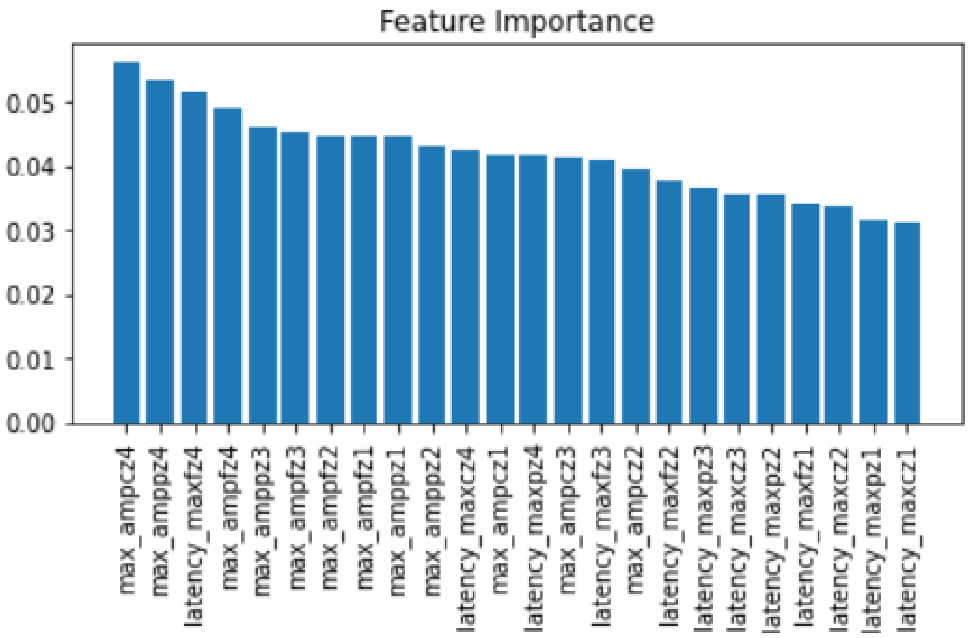

Figure 18: Feature importance for a group of 54 normal subject

for a group of 54 schizophrenia subject

```
Feature: 0, Score: 0.04753
Feature: 1, Score: 0.04349
Feature: 2, Score: 0.04921
Feature: 3, Score: 0.04413
Feature: 4, Score: 0.04474
Feature: 5, Score: 0.04341
Feature: 6, Score: 0.04602
Feature: 7, Score: 0.04249
Feature: 8, Score: 0.04537
Feature: 9, Score: 0.04846
Feature: 10, Score: 0.04608
Feature: 11, Score: 0.05200
Feature: 12, Score: 0.03451
Feature: 13, Score: 0.03186
Feature: 14, Score: 0.03384
Feature: 15, Score: 0.03749
Feature: 16, Score: 0.03581
Feature: 17, Score: 0.04021
Feature: 18, Score: 0.04144
Feature: 19, Score: 0.03607
Feature: 20, Score: 0.03595
Feature: 21, Score: 0.03935
Feature: 22, Score: 0.03957
Feature: 23, Score: 0.04098
```

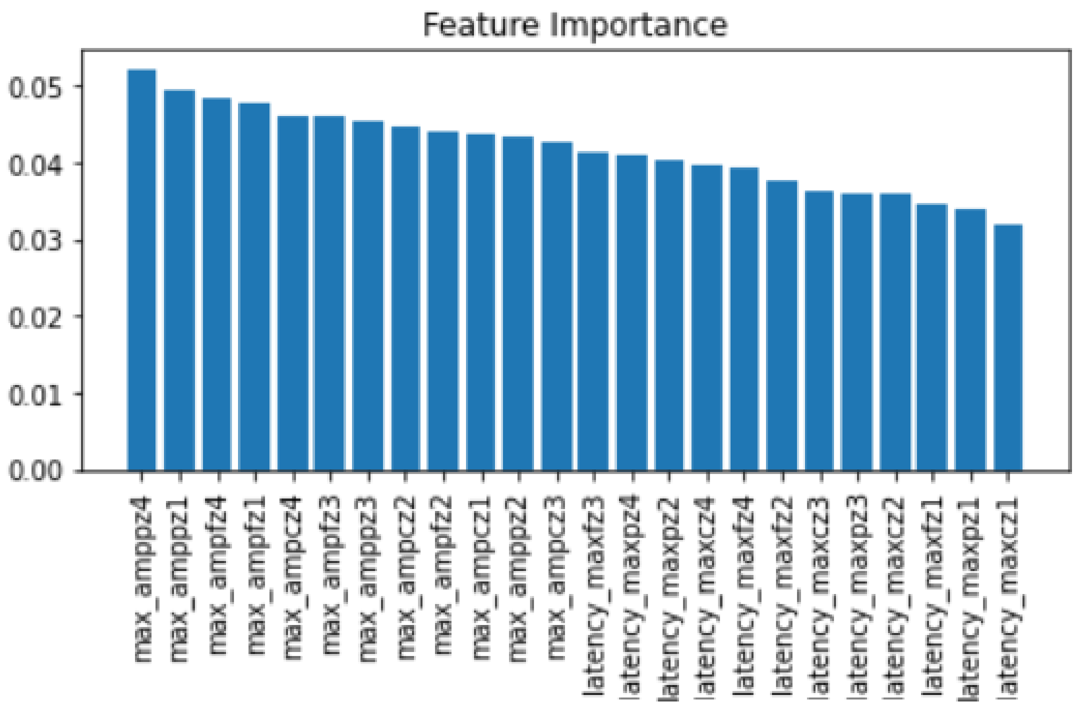

Figure 19: Feature importance for a group of 54 schizophrenia subject

4.3.2 Feature importance connectivity feature

For connectivity features, top 15 most important features are calculated for respective group of 54 normal and schizophrenia subject. The connectivity features are named with respective name of network measures. For example, clustering coefficient is named as clustering_coe and betweenness bin is named as betweennessbin and so on. The number in the feature name represents the feature value for respective channels. For ex. we selected 64 channels for obtaining correlation matrices for each trials in the data and then we retrieved connectivity features from it using brain connectivity toolbox of MATLAB. Thus, for each network measures, we got 64 values and that values are further named by creating features such as clustering_coe_1, clustering_coe_2, and so on till clustering_coe_64. This is done for each respective connectivity features. While creating a dataframe for machine learning part from all this values, trials are arranged at row and each feature values such as clustering_coe_1, clustering_coe_2, etc. are arranged as columns.

for a group of 54 normal subject

```
betweennessbin_62    0.008936
clustering_coe_13    0.008229
clustering_coe_19    0.008084
clustering_coe_30    0.008010
betweennessbin_57    0.007596
clustering_coe_31    0.007559
clustering_coe_32    0.007466
clustering_coe_4     0.007411
clustering_coe_63    0.007385
clustering_coe_60    0.007317
betweennessbin_47    0.007303
betweennessbin_61    0.007280
betweennessbin_64    0.007249
betweennessbin_28    0.007176
betweennessbin_25    0.007136
dtype: float64
```

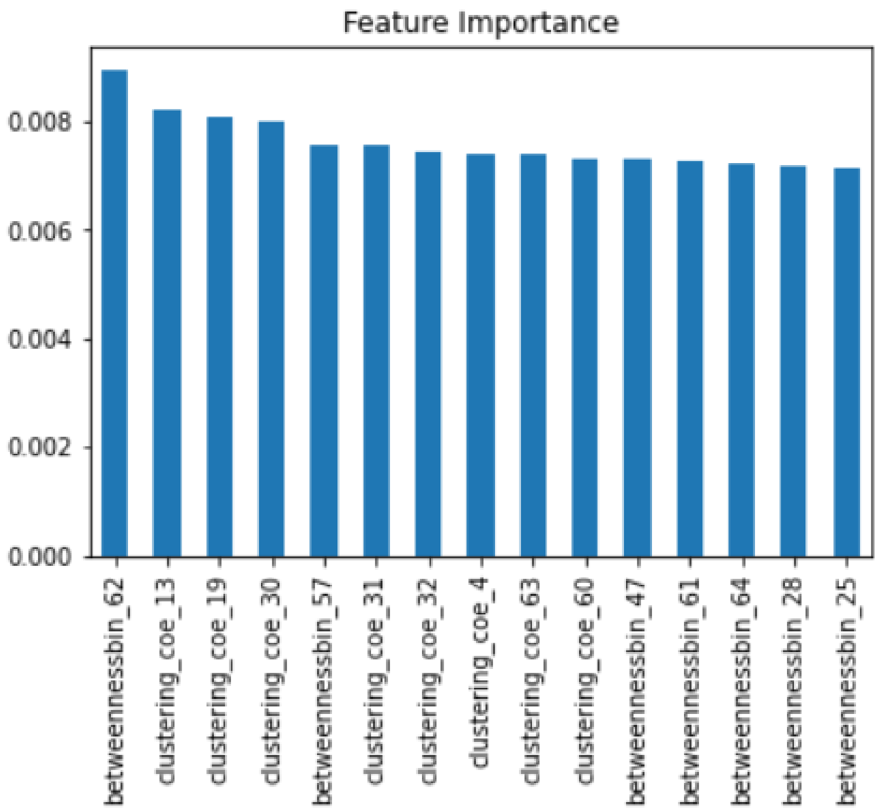

Figure 20: Feature importance for a group of 54 normal subject

for a group of 54 schizophrenia subject

```
clustering_coe_19    0.009927
clustering_coe_31    0.009299
clustering_coe_30    0.008862
clustering_coe_32    0.008369
clustering_coe_24    0.007942
clustering_coe_63    0.007938
clustering_coe_61    0.007913
clustering_coe_59    0.007615
clustering_coe_29    0.007507
clustering_coe_62    0.007500
clustering_coe_56    0.007469
clustering_coe_14    0.007361
betweennessbin_63    0.007269
betweennessbin_53    0.007264
clustering_coe_28    0.007247
dtype: float64
```

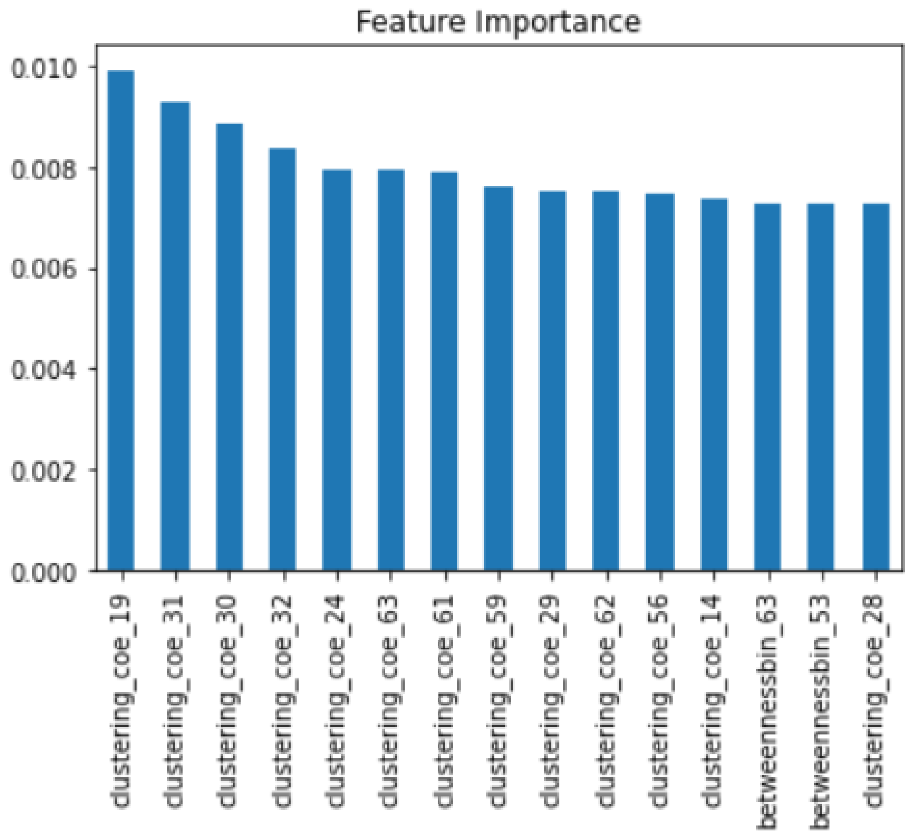

Figure 21: Feature importance for a group of 54 schizophrenia subject

5 Discussion

5.1 Performance with two feature sets

In this thesis, experiments are conducted for single-trial classification of target and distractor from the EEG data. From the experiments, It is observed that connectivity features are giving better accuracy than ERP features in group analysis. When the experiment is run on the group of 10 normal and 10 schizophrenia subject, then ERP features are better than connectivity features for single-trial classification. Also with the single normal and schizophrenia subject, ERP features are giving better accuracy than connectivity features. But, for group of 54 normal and schizophrenia subjects are giving increased accuracy using connectivity features in comparison with ERP features.

During the experiments, We obtained a model accuracy, sensitivity, and specificity. These terms are explained below:

We can use precision, recall, accuracy, and F metrics for the evaluation of binary classification models. In binary classification, the model only has two options to choose from: either the observation belongs to the class or not and the model can be either correct or not so there are four portions of the confusion matrix. We need to calculate metrics for each category if the model has multiple categories as a prediction options[26].

Definitions:

True positive (TP)

These are cases in which we predicted yes for ex. it is a target trial, and it is a target trial.

false positive (FP)

These are cases in which We predicted it is a target trial, but they are not actually target trial. It is also known as a "Type I error".

True negative (TN)

These are cases where we predicted it is not a target trial and it is not a target.

False negative (FN)

These are cases where we predicted it is not a target trial but they are actually target. It is also called as a "Type II error".

Accuracy

The accuracy is the ability of model to differentiate the classes correctly. To estimate the accuracy, we should calculate the proportion of true positive and true negative in all evaluated cases. Below figure shows equation for calculating accuracy metrics:

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

Figure 22: The equations for calculating accuracy[18]

Sensitivity

Sensitivity is the metric which evaluates the model's ability to predict true positives of each available category. To estimate sensitivity, we should calculate the proportion of true positives. Below figure shows equation for calculating Sensitivity metrics:

$$Sensitivity = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

Figure 23: The equations for calculating sensitivity[22]

Specificity

Specificity is the metrics that evaluates a model's ability to predict true negatives of each available category. To estimate specificity, we should calculate the proportion of true negative. Below figure shows equation for calculating specificity metrics:

$$Specificity = \frac{True\ Negative}{True\ Negative + False\ Positive}$$

Figure 24: The equations for calculating specificity[22]

5.2 Feature Importance

5.2.1 Feature importance ERP Feature

Based on the feature importance figures shown in result section i.e. figure 18 & 19, there is a general trend that amplitude feature seems to be more important than latency features for both groups of 54 normal and schizophrenia subjects.

5.2.2 Feature importance connectivity feature

Based on the feature importance figures shown in result section i.e. figure 20 & 21,there is a general trend of the cluster coefficient to be the most important feature for both groups of 54 normal and schizophrenia subjects..

5.3 Limitation

In this thesis, only binary classification between target and distractor trial class is done using ERP and connectivity feature sets. However, there exists a standard trial also with frequency 1000Hz in the data. We can make use of multi-class classification algorithm for this purpose.

5.4 Future work

We can make use of a multi-layer neural network for single-trial classification of EEG data. We can use a multiclass classification to classify standard, target, and distractor trial from EEG data. We can also use other network measures such as network motifs, Participation coefficient, etc. in the connectivity features to determine whether this features are giving different accuracy for classification.

In this experiments, correlation based approach is used to extract the connectivity matrices, but there are other ways as well such as mutual correlation, directed connectivity metrics such as transfer entropy, granger causality, etc [12]. Effective connectivity can be obtained using dynamic causal modelling [11] [5]. However, this framework is suited for small network (6 nodes). Although recent work has developed whole brain dynamic causal modelling but these developments are limited to fMRI data.

6 Conclusion

In this thesis, we implemented a machine learning techniques for single-trial classification of target and distractor trials from the EEG data of normal and schizophrenia subjects. The model proposed in this thesis consists of 2 phases: First phase was a ERP and connectivity feature extraction from the data and second phase was to perform simultaneous processing of feature selection, feature preprocessing, model selection, and hyperparameter tuning using nested cross validation. For connectivity feature extraction, we first created correlation matrices for each trials of data and extracted connectivity features from correlation matrices using brain connectivity toolbox of MATLAB. For extracting ERP features, we have calculated positive and negative peaks for specific time intervals and extracted amplitude and latency of that peak. This feature extraction is described more briefly in feature detection subsection of method section. All feature extraction tasks are performed in MATLAB. Using nested cross validation, We implemented the logistic regression, Decision tree, Random forest, K-nearest neighbor, and SVM classifier on connectivity features and ERP features for the single-trial classification. In this experiments, we got the model accuracy, specificity, and sensitivity which we used for comparing the results between ERP features and connectivity features.

In this thesis, three use-cases of different subject types are developed for comparing accuracy of model to determine whether connectivity features are better/worse than ERP features for single-trial classification. The three usecases on which the experiments were run are: training a model with single normal and schizophrenia subject, group of 10 normal and schizophrenia subjects, and group of 54 normal and schizophrenia subjects. From this experiments we can conclude that, model is giving higher accuracy with connectivity features than ERP features for the group of all 54 normal and schizophrenia subject. Thus, connectivity features are better than ERP features for single-trial classification of target and distractor trials for this dataset.

A Appendix

We measured the performance of our classifiers for ERP features and connectivity features. We calculated the performance for single normal subject, single schizophrenia subject, group of 10 normal subject, and group of 10 schizophrenia subject. For measuring the performances of classifiers, we calculated accuracy, sensitivity, and specificity using nested cross validation. The performances for different subject types with ERP and connectivity features is as shown in below tables.

Performance with ERP features

Performances for single subject

Table 5: Result with ERP features for single subjects

Classifier	Performances	Single normal subject	Single schizophrenia subject
Logistic Regression	Accuracy	0.6571 (0.137)	0.5375 (0.137)
	sensitivity	0.65 (0.2)	0.5333 (0.137)
	specificity	0.6583 (0.2915)	0.55 (0.137)
Random Forest Classifier	Accuracy	0.6839 (0.1429)	0.5785 (0.137)
	sensitivity	0.725 (0.1952)	0.4666 (0.137)
	specificity	0.6333 (0.2304)	0.6916 (0.137)
Decision tree Classifier	Accuracy	0.55 (0.1332)	0.4339 (0.137)
	sensitivity	0.55 (0.225)	0.3416 (0.137)
	specificity	0.5666 (0.1979)	0.5333 (0.137)
KNeighbors Classifier	Accuracy	0.5928 (0.15)	0.5982 (0.137)
	sensitivity	0.55 (0.2358)	0.625 (0.137)
	specificity	0.6416 (0.225)	0.5916 (0.137)
SVM Classifier	Accuracy	0.6446 (0.119)	0.5232 (0.137)
	sensitivity	0.6 (0.223)	0.5083 (0.137)
	specificity	0.6916 (0.1600)	0.5416 (0.137)

mean and standard deviation of accuracy from all classifiers

Table 6: Mean and standard deviation of accuracy		
Subject Types	Performances	mean (standard deviation)
For single normal subject	Accuracy	0.6839 (0.1267)
	sensitivity	0.675 (0.2512)
	specificity	0.6833 (0.2512)
For single schizophrenia subject	Accuracy	0.6392 (0.1584)
	sensitivity	0.5833 (0.2527)
	specificity	0.7 (0.3172)

Above table shows the mean and standard deviation from all classifiers using ERP features for single subject.

Performances for group of 10 subject ERP features

Table 7: Result with ERP features for group of 10 subjects

Classifier	Performances	group of 10 normal subject	group of 10 schizophrenia subject
Logistic Regression	Accuracy	0.5568 (0.074)	0.5340 (0.0639)
	sensitivity	0.5553 (0.101)	0.5410 (0.0471)
	specificity	0.5587 (0.0852)	0.5261 (0.1034)
Random Forest Classifier	Accuracy	0.5716 (0.0496)	0.5176 (0.0473)
	sensitivity	0.5785 (0.0644)	0.5256 (0.0512)
	specificity	0.5645 (0.0636)	0.5097 (0.0661)
Decision tree Classifier	Accuracy	0.5759 (0.0580)	0.5436 (0.0489)
	sensitivity	0.5756 (0.0602)	0.5757 (0.0576)
	specificity	0.5762 (0.1033)	0.5095 (0.1000)
KNeighbors Classifier	Accuracy	0.5553 (0.0483)	0.5204 (0.0167)
	sensitivity	0.5403 (0.0637)	0.5415 (0.0526)
	specificity	0.5706 (0.0547)	0.4983 (0.0662)
SVM Classifier	Accuracy	0.5701 (0.0471)	0.5218 (0.0439)
	sensitivity	0.5700 (0.0758)	0.5408 (0.0646)
	specificity	0.5701 (0.0906)	0.5011 (0.0761)

mean and standard deviation of accuracy from all classifiers

Table 8: Mean and standard deviation of accuracy		
Subject Types	Performances	mean (standard deviation)
For group of 10 normal subject	Accuracy	0.5628 (0.0653)
	sensitivity	0.5583 (0.0519)
	specificity	0.5674 (0.0955)
For group of 10 schizophrenia subject	Accuracy	0.5231 (0.0650)
	sensitivity	0.5362 (0.0600)
	specificity	0.5099 (0.1027)

Above table shows mean and standard deviation from all classifiers for group of 10 subject with ERP features.

Performance with connectivity features

Performances for single subject

Table 9: Result with connectivity features for single subjects

Classifier	Performances	Single normal subject	Single schizophrenia subject
Logistic Regression	Accuracy	0.4714 (0.1925)	0.4446 (0.126)
	sensitivity	0.375 (0.3316)	0.45 (0.172)
	specificity	0.5666 (0.3162)	0.45 (0.219)
Random Forest Classifier	Accuracy	0.5089 (0.1322)	0.4946 (0.2010)
	sensitivity	0.475 (0.2358)	0.5583 (0.2614)
	specificity	0.5416 (0.2641)	0.45 (0.2656)
Decision tree Classifier	Accuracy	0.5214 (0.1232)	0.5535 (0.2001)
	sensitivity	0.55 (0.2449)	0.5583 (0.1709)
	specificity	0.4916 (0.3135)	0.5583 (0.3201)
KNeighbors Classifier	Accuracy	0.5464 (0.1523)	0.4267 (0.1515)
	sensitivity	0.50 (0.3)	0.4166 (0.16)
	specificity	0.5916 (0.2783)	0.45 (0.214)
SVM Classifier	Accuracy	0.4964 (0.1455)	0.5285 (0.1888)
	sensitivity	0.425 (0.2839)	0.5666 (0.1870)
	specificity	0.5666 (0.2986)	0.5083 (0.2473)

mean and standard deviation of accuracy from all classifiers

Table 10: Mean and standard deviation from all classifier		
Subject Types	Performances	Result
For single normal subject	Accuracy	0.5589 (0.0931)
	sensitivity	0.45 (0.2358)
	specificity	0.6666 (0.1979)
For single schizophrenia subject	Accuracy	0.4803 (0.1474)
	sensitivity	0.4666 (0.1384)
	specificity	0.5 (0.2656)

Above table shows the mean and standard deviation from all classifiers using connectivity features for single subject.

Performances for group of 10 subject

Table 11: Result with connectivity features for group of 10 subjects			
Classifier	Performances	group of 10 normal subject	group of 10 schizophrenia subject
Logistic Regression	Accuracy	0.5334 (0.0826)	0.5559 (0.0462)
	sensitivity	0.5263 (0.1363)	0.6078 (0.0904)
	specificity	0.5401 (0.0759)	0.5015 (0.1172)
Random Forest Classifier	Accuracy	0.5171 (0.0513)	0.5615 (0.0514)
	sensitivity	0.4679 (0.0992)	0.6267 (0.0513)
	specificity	0.5672 (0.1193)	0.4930 (0.1143)
Decision tree Classifier	Accuracy	0.5435 (0.0859)	0.5641 (0.0494)
	sensitivity	0.5112 (0.1049)	0.5520 (0.1164)
	specificity	0.5757 (0.0905)	0.5773 (0.0880)
KNeighbors Classifier	Accuracy	0.4964 (0.0615)	0.5368 (0.0449)
	sensitivity	0.5033 (0.0882)	0.5701 (0.1046)
	specificity	0.4898 (0.0491)	0.5020 (0.1278)
SVM Classifier	Accuracy	0.5541 (0.0725)	0.5492 (0.0513)
	sensitivity	0.5499 (0.0829)	0.5625 (0.0623)
	specificity	0.5580 (0.0969)	0.5355 (0.1236)

mean of accuracy from all classifiers

Table 12: Mean and standard deviation connectivity features		
Subject Types	Performances	Result
For group of 10 normal subject	Accuracy	0.5052 (0.0634)
	sensitivity	0.4591 (0.0910)
	specificity	0.5520 (0.0943)
For group of 10 schizophrenia subject	Accuracy	0.5614 (0.0479)
	sensitivity	0.6188 (0.1014)
	specificity	0.5020 (0.1296)

Here, we obtained the result from the single normal and schizophrenia subject and group of 10 normal and schizophrenia subject. From this subject types we do not observed a improved accuracy using connectivity feature than ERP feature. But as mentioned in result section, if we run the experiment for all 54 normal and schizophrenia subject, we got better accuracy with connectivity feature as compared to ERP feature. Thus, connectivity features are better than ERP features for classifying single-trial EEG data.

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